Polar Measurements on a Small Turntable: How Well Can the Rotation Center Be Corrected?

A free-field simulation study of a horizontal three-way monitor, with a boundary-element check

Technical report — simulation code and data:
https://github.com/Arnwald/turntable-rotation-center-study

Abstract

A polar measurement should rotate the loudspeaker about a reference point on its baffle. The turntables available to small manufacturers and hobbyists are usually too small for that: the cabinet ends up centered on the turntable, the rotation axis lies well behind the baffle and, for an asymmetric layout, off to the side. The microphone is then farther away and in another direction than intended. A purely geometric correction — each driver treated as a point source with its own directivity, measurements re-propagated in 3D to the intended distance and their powers interpolated to the intended direction — can rebuild the intended polar data. We test it in free field on a simulated horizontal three-way monitor (8-inch woofer, 3-inch dome midrange, 1.1-inch dome tweeter, 440 × 290 × 290 mm closed box) whose cabinet diffraction follows the Distributed Edge Dipole model, for a horizontal orbit from −180° to 180° in 10° steps at 1 m and 2 m. Without diffraction the correction is close to exact (0.10 dB RMS on the summed loudspeaker at 1 m). With diffraction, about 0.24 dB RMS remains on the summed loudspeaker at 1 m (2.45 dB without correction), concentrated behind the cabinet at mid and high frequencies. The Predicted In-Room Response computed from the corrected data stays within 0.12 dB (1 m) and 0.08 dB (2 m) of the rigorous one, instead of 1.02 dB and 0.48 dB uncorrected. The lateral offset of the microphone matters for the individual curves: omitted, or entered with the wrong sign, it leaves four to seven times the residual error, although the spatial averages hardly notice. A boundary-element solution of a two-way variant in the same cabinet, with extended rigid radiators and the whole cabinet, confirms these results below about 3 kHz but shows more diffraction above: the rigid dome tweeter radiates wider than its aperture model, the far edge of the baffle adds fine interference ripples, and with 10° steps the corrected PIR deviates by up to 0.55 dB in narrow bands at 1 m (0.32 dB at 2 m). That residual is mostly the angular step: with 2.5° steps the PIR comes within 0.15 dB at 1 m and 0.10 dB at 2 m. Interpolating powers rather than complex responses halves the residual of the individual curves with 10° steps. The vertical orbit, measured with the cabinet on its side, is corrected the same way. Rebuilt as from a 2 m circle, measurements taken at 1 m keep the cabinet's near field: the diffraction pattern and the low-frequency acoustic center, which no point-source model moves to another distance.

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1 Introduction

The off-axis behavior of a loudspeaker is measured by turning it in front of a fixed microphone. Standards such as CTA-2034-A place the center of rotation on the baffle, at the reference point (usually the tweeter axis), so that the microphone moves on a circle of constant radius around it. With a large turntable, or a rig that holds the cabinet with its baffle over the axis, that is easy. The turntables most small manufacturers and hobbyists can afford (a few hundred millimeters across) cannot carry a cabinet that overhangs by 30 to 45 cm: the cabinet is centered on the turntable, and the rotation axis passes through the middle of the box (Figure 4).

Two things then go wrong. Off axis, the microphone is no longer at the intended distance from the drivers: farther away behind the cabinet, so the off-axis responses are too low and too late. And the direction in which each driver is seen is no longer the nominal angle. When the drivers are not on the cabinet center line — a horizontal monitor with its woofer beside the midrange-tweeter

column, as here — the microphone has to be moved sideways to face the reference point at 0°, and the errors become asymmetric.

A correction can be built from geometry alone. Around each driver, a point source with an arbitrary directivity is assumed; every measurement is moved to the intended distance (a delay and a gain) and the response in the intended direction is interpolated between the two measurements that bracket it. The method needs four numbers per plane: where the rotation axis was relative to the baffle (depth and lateral offset), where each driver is relative to the reference axis, and the microphone distance; the drivers' heights relative to the microphone make the distances exact in 3D. The same model can rebuild the data as seen from a larger circle than the one measured on (CTA-2034-A measures at 2 m), which Section 6.7 tests too.

This report checks how well that works. The setting is deliberately idealized: free field (no room, no gating), a loudspeaker simulated from its drivers' datasheets, and a diffraction model for the cabinet edges, which is the open question — the edges radiate from places that are not the driver, so a point-source correction cannot be exact for them. We compare, at 1 m and 2 m, the rigorous measurement (axis on the baffle), the raw measurement with the cabinet centered on the turntable, and the same measurement after correction, driver by driver, for a summed three-way loudspeaker and for the CTA-2034-A spatial averages, Predicted In-Room Response included. We also look at what happens when the correction parameters are wrong, which is the practical risk.

2 The simulated loudspeaker

2.1 Cabinet and layout

The loudspeaker is a horizontal three-way monitor in a closed box of $W \times H \times D = 440 \times 290 \times 290$ mm (24 L net), Figure 1. The woofer is on the left; the midrange and the tweeter are stacked on the right, tweeter on top. All positions are expressed in the *speaker frame*: origin at the center of the front baffle, x to the right as seen from the listening position, y up, z normal to the baffle toward the listener. The cabinet occupies $-D \le z \le 0$ and every driver sits on the baffle plane $z = 0$ (Table 1). The *reference point* $\mathbf{R}$ is on the baffle, on the vertical line through the midrange and tweeter centers, halfway between them; the *reference axis* is the baffle normal through $\mathbf{R}$. The microphone is placed on that axis, at the height of $\mathbf{R}$.

2.2 Driver responses from the datasheets

Each driver is described by its on-axis half-space response $H_i(f)$: complex pressure at $r_{\mathrm{ref}} = 1$ m for 2.83 V, in an infinite baffle. The SPL curves $L_i(f)$ of the three datasheets are vector graphics; they were extracted point by point from the PDF files rather than traced by eye (the CSV files are in the repository). Each curve is taken as the half-space response, interpolated linearly in $\log f$ and extended outside its published range with fixed slopes s_{lo} and s_{hi} (dB per octave):

$$L(f) = L(f_{\min}) + s_{\mathrm{lo}} \log_2 \frac{f}{f_{\min}} \quad (f < f_{\min}), \qquad L(f) = L(f_{\max}) + s_{\mathrm{hi}} \log_2 \frac{f}{f_{\max}} \quad (f > f_{\max}), \tag{1}$$

with $s_{\mathrm{lo}} = +12$ for all three drivers and $s_{\mathrm{hi}} = -24$ (woofer) or -12 (domes). The magnitude $|H_i| = p_{\mathrm{ref}}\, 10^{L_i/20}$ ($p_{\mathrm{ref}} = 20$ µPa) is given a minimum phase with the folded real cepstrum: on an

Table 1: Driver positions in the speaker frame and radiator models.

	x (mm)	y (mm)	radiator
Woofer, Dayton Audio SIG225-4	−85	0	flat piston, $a = \sqrt{S_d/\pi} = 83.8$ mm
Midrange, HiVi DM-7600	120	−48.5	equivalent cap, $a = 60$ mm, $h = 17.5$ mm
Tweeter, SICA LP 90.28/N92	120	81.5	spherical cap, $a = 14$ mm, $h = 7.5$ mm
Reference point $\mathbf{R}$	120	16.5	—

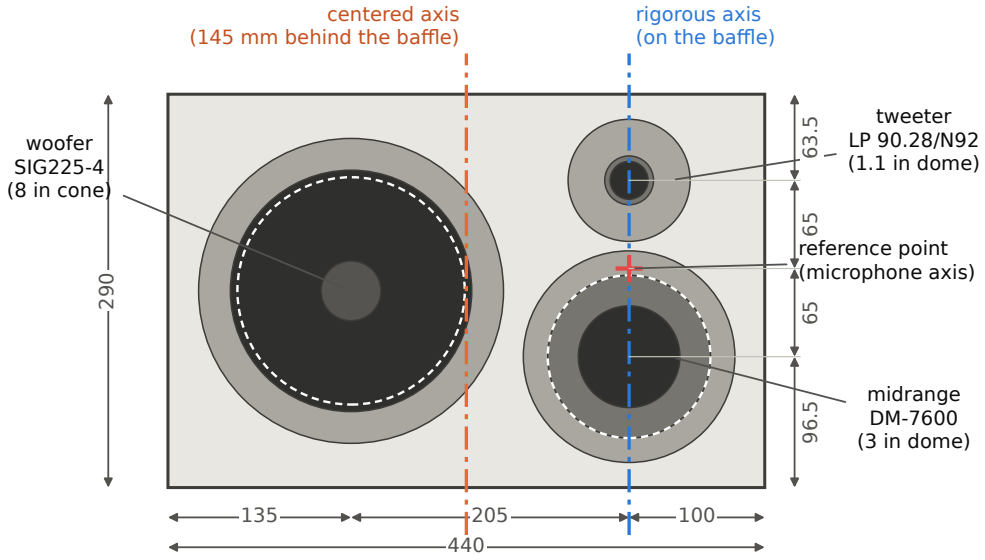


Figure 1: Front view of the simulated monitor (dimensions in mm). Dashed white circles: radiators used in the model (piston of the woofer's S_d, equivalent radiating cap of the midrange). Blue: rotation axis of the rigorous setup, on the baffle through midrange and tweeter. Orange: rotation axis when the cabinet is centered on the turntable, 145 mm behind the baffle.

N-point DFT grid,

$$c[n] = \mathrm{IDFT}\{\ln|H[k]|\}, \qquad \hat{c}[n] = \begin{cases} c[n] & n = 0,\ n = N/2 \\ 2c[n] & 0 < n < N/2 \\ 0 & N/2 < n < N \end{cases}, \qquad H_{\min}[k] = \exp\left(\mathrm{DFT}\{\hat{c}\}[k]\right), \tag{2}$$

evaluated on a grid eight times finer than the simulation grid to avoid cepstral aliasing.

Woofer in its closed box. The SIG225-4 datasheet curve is close to an infinite-baffle response at low frequency (its low end follows $f_s = 40\,\mathrm{Hz}$, $Q_{ts} = 0.43$). It is moved into the 24 L box with the ratio of the two second-order high-pass functions,

$$H_{\mathrm{woofer}}(f) = H_{\mathrm{ds}}(f)\,\frac{\mathcal{H}(f; f_c, Q_{tc})}{\mathcal{H}(f; f_s, Q_{ts})}, \qquad \mathcal{H}(f; f_0, Q) = \frac{s^2}{s^2 + s/Q + 1},\ s = \mathrm{j}\frac{f}{f_0}, \tag{3}$$

$$\alpha = \frac{V_{as}}{V_b} = \frac{29.7}{24}, \qquad f_c = f_s\sqrt{1+\alpha} = 59.8\ \mathrm{Hz}, \qquad Q_{tc} = Q_{ts}\sqrt{1+\alpha} = 0.64 \tag{4}$$

(lossless box). Both high-pass functions are minimum phase, so the product stays minimum phase. As a check, the Thiele–Small parameters give a mass-controlled sensitivity

$$p = \frac{\rho_0\, S_d\, Bl\, V}{2\pi r\, R_e\, M_{ms}} = 0.75\ \mathrm{Pa} \quad (\rho_0 = 1.204\ \mathrm{kg/m^3}), \tag{5}$$

that is 91.5 dB at 1 m for the datasheet's 91.4 dB.

Band limit. All responses are multiplied by a raised-cosine taper from 20.5 to 23.5 kHz, which keeps the impulse responses band-limited at the 48 kHz sample rate.

Figure 2 compares models and datasheets.

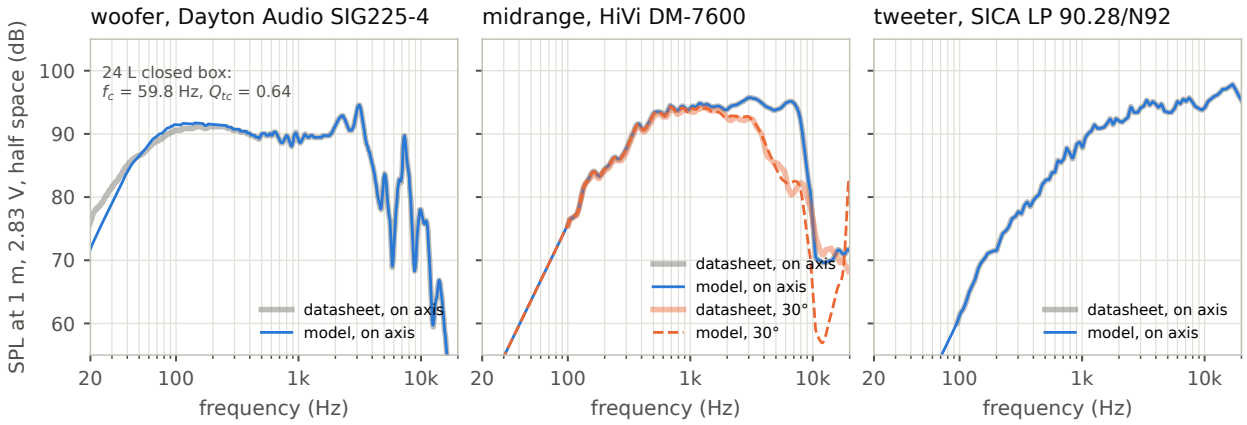


Figure 2: Driver models against the datasheet curves (half space, 1 m, 2.83 V). The woofer is shown in its 24 L closed box. For the midrange, the datasheet also gives the 30° response, which the equivalent radiating cap reproduces to within 0.8 dB RMS between 1.5 and 8 kHz.

2.3 Radiators

The directivities are the classical far-field ones [4], normalized to 1 on the driver's own axis (θ is the angle from that axis, $k = 2\pi f/c$, $c = 343\,\mathrm{m/s}$).

Cone: rigid flat piston of radius $a = \sqrt{S_d/\pi}$ in an infinite baffle,

$$D_{\mathrm{p}}(\theta, k) = \frac{2J_1(ka\sin\theta)}{ka\sin\theta}. \tag{6}$$

Dome: rigid convex spherical cap of base radius a and height h, moving axially. Its far field is the Rayleigh integral over the projected disc, each ring of radius r being advanced by the cap height $z(r)$:

$$A(\theta, k) = \frac{2}{a^2}\int_0^a J_0(kr\sin\theta)\, e^{\mathrm{j}kz(r)\cos\theta}\, r\,\mathrm{d}r, \qquad z(r) = \sqrt{R_s^2 - r^2} - (R_s - h), \quad R_s = \frac{a^2 + h^2}{2h}, \tag{7}$$

$$D_{\mathrm{d}}(\theta, k) = \frac{A(\theta, k)}{A(0, k)}. \tag{8}$$

D_{d} is complex: the protruding center moves the apparent source off axis. With $h = 0$ it is the piston. The integral is evaluated with 96-point Gauss–Legendre quadrature.

The midrange's equivalent cap. The DM-7600 sits in a horn-shaped aluminum faceplate. A bare 75 mm dome is far wider than the datasheet's 30° curve (3 dB down at 5 kHz where the datasheet shows 8 dB). Fitting (a, h) of Eq. (7) to the datasheet difference $L_{30^\circ} - L_{0^\circ}$ between 1.5 and 8 kHz gives $a = 60\,\mathrm{mm}$, $h = 17.5\,\mathrm{mm}$ (0.83 dB RMS): a 120 mm aperture, the diameter of the faceplate's mouth (the baffle cutout is 120.5 mm). That equivalent cap is used for the midrange. The tweeter, with no off-axis data, uses its own geometry: a 28 mm silk dome, assumed 7.5 mm high.

2.4 Cabinet diffraction: the Distributed Edge Dipole model

The edges of the front baffle are modelled with the Distributed Edge Dipole (DED) approach of Urban *et al.* [1], a refinement of the edge-source picture of Vanderkooy [3]: a line of dipoles along the edges, driven by the pressure the driver produces there. In the form used here, extended to a source of finite size, driver i produces at the microphone position $\mathbf{M}$

$$p_i(\mathbf{M}, f) = H_i(f)\,\left[P_{\mathrm{dir},i}(\mathbf{M}, f) + P_{\mathrm{edge},i}(\mathbf{M}, f)\right], \tag{9}$$

$$P_{\mathrm{dir},i} = D_i(\theta_i, k)\,\frac{1+\cos\theta_i}{2}\,\frac{r_{\mathrm{ref}}}{r_i}\,e^{-\mathrm{j}kr_i}, \qquad r_i = |\mathbf{M}-\mathbf{S}_i|, \quad \cos\theta_i = \frac{\mathbf{n}\cdot(\mathbf{M}-\mathbf{S}_i)}{r_i}, \tag{10}$$

$$P_{\mathrm{edge},i} = \sum_e Q_{i,e}(f)\,F_e(\mathbf{M})\,\frac{r_{\mathrm{ref}}}{r_{2,e}}\,e^{-\mathrm{j}kr_{2,e}}, \qquad r_{2,e} = |\mathbf{M}-\mathbf{E}_e|, \quad F_e(\mathbf{M}) = -\frac{\mathbf{n}\cdot(\mathbf{M}-\mathbf{E}_e)}{r_{2,e}}, \tag{11}$$

where $\mathbf{S}_i$ is the driver center, $\mathbf{n} = \mathbf{e}_z$ the baffle normal, D_i the radiator directivity, and $\mathbf{E}_e$, Δl_e the midpoints and lengths of the elements along the four edges of the baffle (2 mm elements, 730 in all). The factor $(1+\cos\theta)/2$ is the obliquity of the direct field: 1 on axis, $\frac{1}{2}$ in the baffle plane, 0 behind; F_e is the edge dipole, zero in the baffle plane and changing sign behind it. The strength of edge element e is

$$Q_{i,e}(f) = C_i\,\Delta l_e\,\frac{G_i(\rho_{i,e}, f)}{A_i(0,k)}, \qquad G_i(\rho, f) = \frac{1}{S_i}\iint_{S_i}\frac{1+\cos\theta_s}{2}\,\frac{e^{-\mathrm{j}kr_1}}{r_1}\,\mathrm{d}S, \tag{12}$$

where the integral runs over the radiating surface (the projected disc, each point raised by the cap height $z(r)$ for a dome), r_1 is the distance from the surface point to the edge point, $\cos\theta_s = -z(r)/r_1$ the obliquity toward it, $\rho_{i,e} = |\mathbf{E}_e - \mathbf{S}_i|$, and $A_i(0,k)$ (Eq. 7; 1 for a piston) refers the edge drive to the on-axis level that H_i describes. The constant

$$C_i = \frac{1}{2\sum_e \Delta l_e\,G_i(\rho_{i,e}, 0)} \tag{13}$$

makes $P_{\mathrm{dir}} + P_{\mathrm{edge}} = \frac{1}{2}$ on axis in the far field at low frequency: the cabinet radiates 6 dB below the half-space response at low frequency (full-space radiation) and reaches it at high frequency — the baffle step.

All paths keep their true length: the edge waves arrive $(r_1 + r_2 - r)/c$ after the direct sound, *at the actual microphone position*. That finite-distance geometry is exactly what a point-source correction cannot reproduce, and the reason for using the DED model here. As in the DED formulation, only the edges of the face that carries the drivers are modelled; behind the baffle the field is the sum of the edge dipoles plus the vanishing direct term.

Numerics. $G_i(\rho, f)\,e^{\mathrm{j}k\rho}$ varies slowly with ρ and f once the common delay ρ/c is removed. It is tabulated on a 1 mm grid of ρ and a 50 Hz grid of f, with a Gauss–Legendre rule in r and a midpoint rule in azimuth over half a turn (the integrand is even), sized for the phase spread $2k_{\max}a$ across the surface ($n_r = \lceil 0.7 \cdot 2k_{\max}a + 16\rceil$, $n_\varphi = \lceil 0.5 \cdot 2k_{\max}a + 16\rceil$); it is then interpolated linearly in ρ and with cubic splines in f. On each driver's axis at 30 m, $P_{\mathrm{dir}} + P_{\mathrm{edge}}$ is at -5.97 dB at 20 Hz for all three drivers, and within 0.3 dB of 0 dB at 10 kHz.

2.5 An illustrative three-way crossover

To look at the summed loudspeaker, each driver is equalized to an acoustic target on the reference axis at 1 m (rigorous setup):

$$T_{\mathrm{woofer}} = \mathrm{LR}_4^{\mathrm{LP}}(450)\,\mathrm{B}_2^{\mathrm{HP}}(40), \quad T_{\mathrm{mid}} = \mathrm{LR}_4^{\mathrm{HP}}(450)\,\mathrm{LR}_4^{\mathrm{LP}}(3000), \quad T_{\mathrm{tweeter}} = \mathrm{LR}_4^{\mathrm{HP}}(3000), \tag{14}$$

with $\mathrm{B}_2^{\mathrm{LP}}(f_0) = 1/(s^2 + \sqrt{2}s + 1)$, $\mathrm{B}_2^{\mathrm{HP}} = s^2/(s^2 + \sqrt{2}s + 1)$, $s = \mathrm{j}f/f_0$, and $\mathrm{LR}_4 = \mathrm{B}_2^2$. The filter of driver i is the minimum-phase filter (Eq. 2) with magnitude $L\,|T_i|/S_i$, where S_i is the 1/6-octave smoothed on-axis magnitude of the driver and L corresponds to 86 dB SPL. The summed response is $p_{\mathrm{sys}} = \sum_i X_i\,p_i$. The same filters are used in every case: only the measurements change. The design is not optimized (Figure 3, right) and does not need to be.

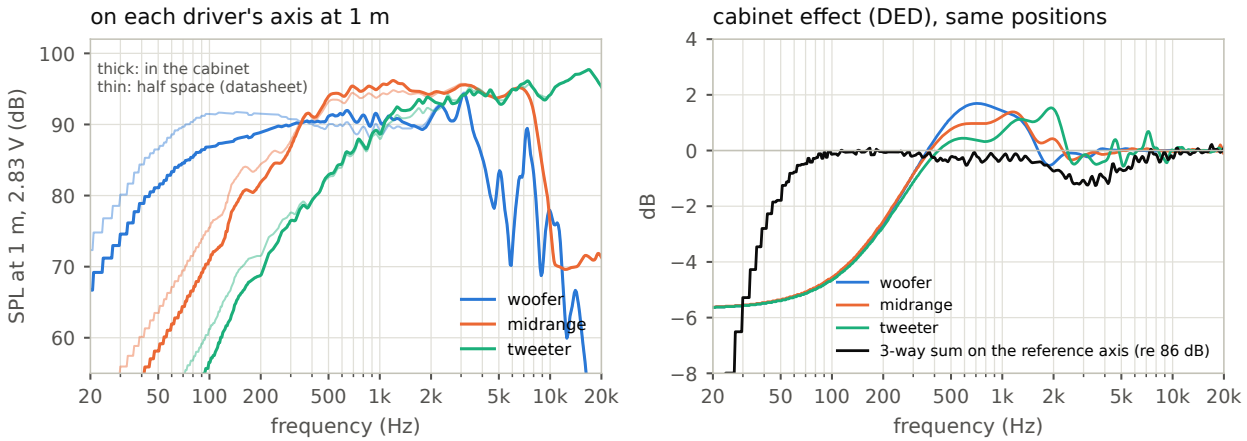


Figure 3: Left: each driver on its own axis at 1 m, in the cabinet (thick) and in half space (thin). Right: their ratio, the cabinet effect according to the DED model (baffle step and edge ripple), and the illustrative three-way sum on the reference axis.

3 Measurement geometry

3.1 Turntable kinematics

The turntable axis is vertical, at $\mathbf{A} = (a_x, \cdot, a_z)$ in the speaker frame. The microphone is fixed in the room; at 0° it is on the reference axis, at distance d from the baffle: $\mathbf{M}_0 = (x_R,\, y_R,\, d)$. Turning the loudspeaker by θ moves the microphone, seen from the loudspeaker, by $-\theta$ about the axis. With positive angles bringing the microphone toward $+x$ (the midrange/tweeter side):

$$\begin{aligned} x(\theta) &= a_x + (x_R - a_x)\cos\theta + (d - a_z)\sin\theta, \\ z(\theta) &= a_z - (x_R - a_x)\sin\theta + (d - a_z)\cos\theta, \end{aligned} \qquad y(\theta) = y_R. \tag{15}$$

- **Rigorous setup**: $\mathbf{A}$ on the reference point, $(a_x, a_z) = (x_R, 0)$. The microphone stays on the circle of radius d around $\mathbf{R}$: $(x, z) = (x_R + d\sin\theta,\ d\cos\theta)$.
- **Centered setup**: the cabinet is centered on the turntable, $(a_x, a_z) = (0, -D/2)$, i.e. 145 mm behind the baffle and 120 mm to the left of the reference axis. The microphone has been moved 120 mm sideways so that it faces the reference point at 0° (Figures 4 and 5).

3.2 What the raw measurement gets wrong

Seen from a driver at $\mathbf{S}$, the microphone of the centered setup is at a distance $\rho_k = |\mathbf{M}(\theta_k) - \mathbf{S}|$ and in a direction φ_k that differ from the intended ones (Figure 6). Figure 7 and Table 2 give the consequences in the horizontal plane: the raw off-axis responses are up to 2.5–2.8 dB too low and almost 1 ms late behind the cabinet at 1 m, and the direction is wrong by up to 15°. At 2 m the level and direction errors halve; the delay does not (it is set by the axis offset, not by the distance).

Table 2: Range of the geometric errors of the raw measurement (centered setup).

driver	d	level (dB)	delay (ms)	direction (°)
tweeter / midrange	1 m	−2.53 to +0.33	−0.11 to +0.99	−3.4 to +15.4
woofer	1 m	−2.81 to +0.17	−0.06 to +0.94	−5.2 to +14.4
tweeter / midrange	2 m	−1.35 to +0.17	−0.12 to +0.98	−1.8 to +8.2
woofer	2 m	−1.41 to +0.13	−0.09 to +0.95	−2.3 to +7.9

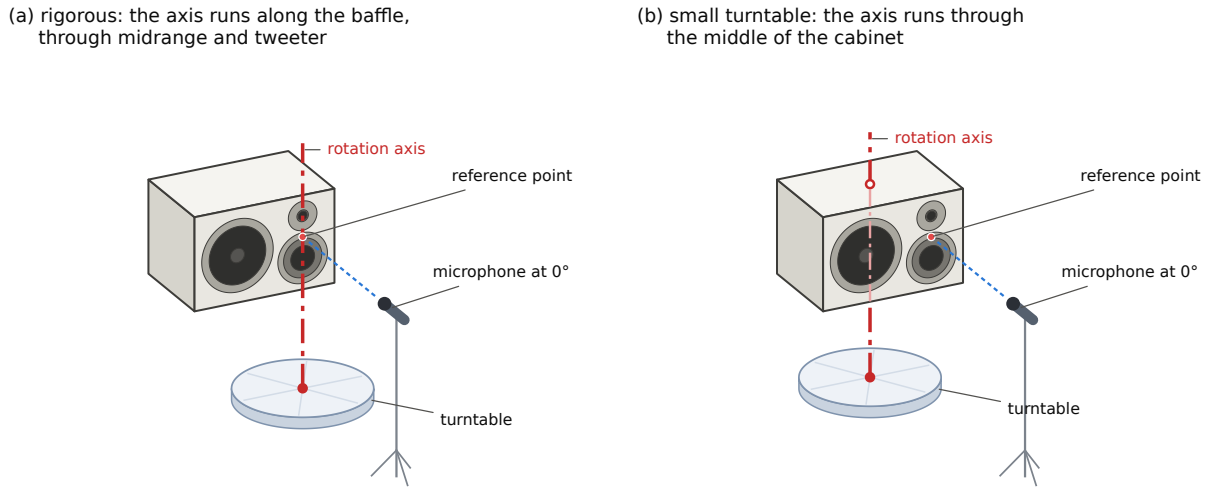


Figure 4: The two setups, with the loudspeaker raised above the disc that stands for the turntable. Dash-dotted red: the rotation axis — (a) along the baffle, through the midrange and the tweeter; (b) through the middle of the cabinet (lighter where it is hidden inside the box). The microphone faces the reference point (red dot) at 0° in both cases.

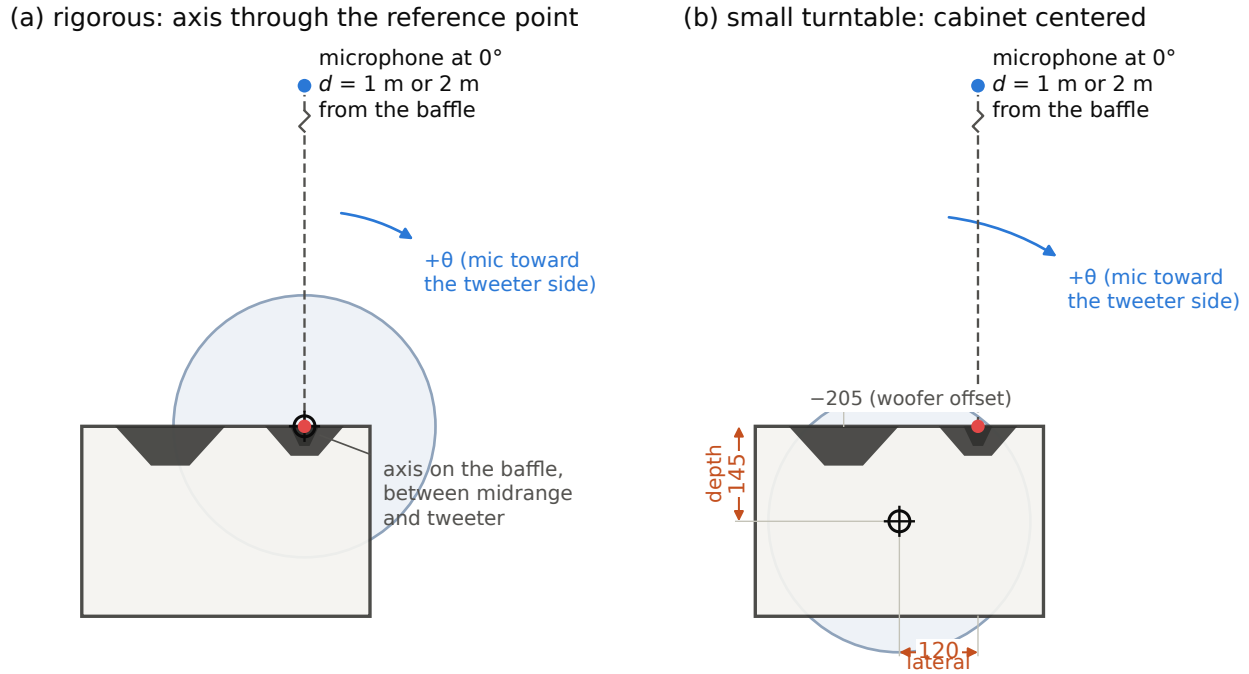


Figure 5: Top views to scale (turntable 400 mm across; microphone distance not to scale). In (b) the correction needs the depth (rotation axis to baffle, 145 mm), the lateral offset (rotation axis to reference axis, +120 mm, positive toward the side the microphone faces at +90°), each driver's offset from the reference axis (−205 mm for the woofer, 0 for midrange and tweeter) and the microphone distance.

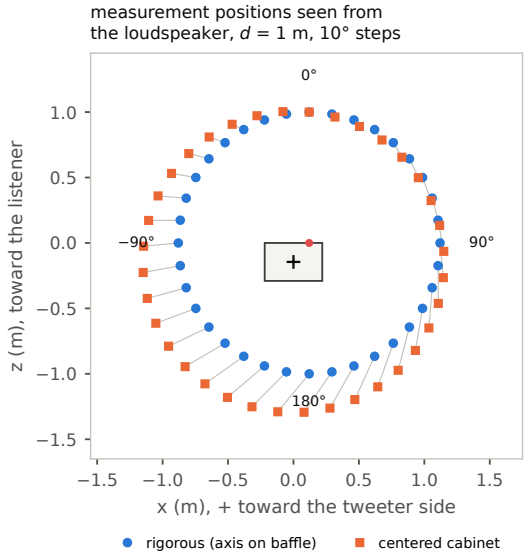


Figure 6: Microphone positions relative to the loudspeaker, to scale, at 1 m. Blue: rigorous setup, a circle of 1 m around the reference point. Orange: centered setup, a circle of 1.15 m around the middle of the cabinet. Gray segments join the positions with the same nominal angle.

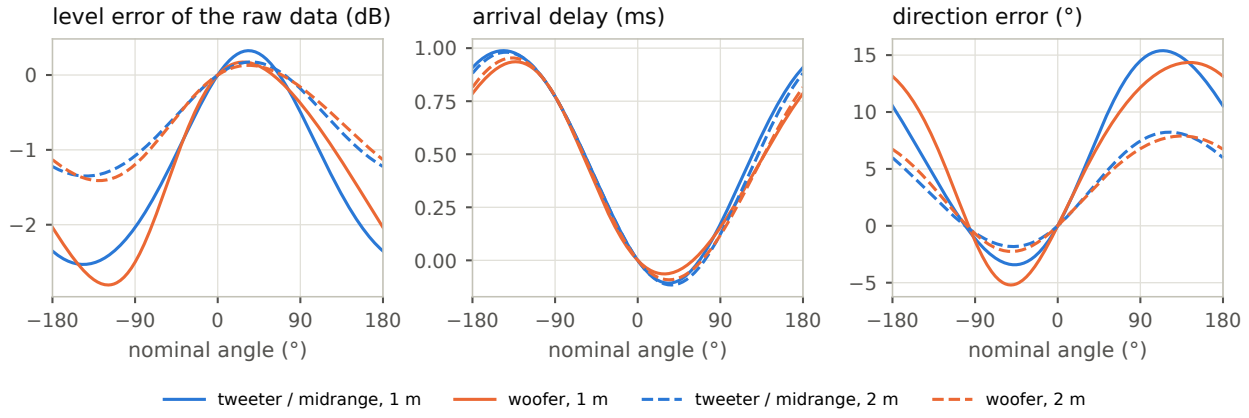


Figure 7: Errors of the raw measurement in the centered setup, seen from the drivers: level ($20 \log_{10}$ of the distance ratio), extra arrival delay and direction error, against the nominal angle.

4 The correction method

The correction rebuilds, for each nominal angle, the response that would have been measured at the ideal position. It works plane by plane and driver by driver.

4.1 Inputs and geometry

In the plane of rotation, take a frame with the rotation axis at the origin, y toward the front and x toward the *+ side* (the side the microphone faces at +90°). The user provides

- the depth δ: from the rotation axis to the baffle (positive when the axis is behind it);
- the lateral offset ℓ: from the rotation axis to the reference axis, positive toward the + side;
- the driver offset o: from the reference axis to the driver, positive toward the + side;
- the microphone distance r_0 from the baffle at 0°;
- the driver's height h above the microphone, out of the plane of rotation.

Then the reference point is $\mathbf{R} = (\ell, \delta, 0)$, the driver $\mathbf{S} = \mathbf{R} + (o, 0, h)$ (third coordinate across the plane), and

$$\text{measured position:} \quad \mathbf{M}_k = \mathcal{R}(\theta_k)\,(\ell,\ \delta + r_0,\ 0), \tag{16}$$

$$\text{ideal position:} \quad \mathbf{M}^*(\theta) = \mathbf{R} + r^*(\sin\theta,\ \cos\theta,\ 0), \tag{17}$$

with the rotation $\mathcal{R}(\theta)(x, y, z) = (x\cos\theta + y\sin\theta,\ -x\sin\theta + y\cos\theta,\ z)$ (positive angles toward the + side) and $r^* = r_0$ unless the results are rebuilt at another distance (Section 4.4). The 0° measurement is on the reference axis, ℓ away from the rotation axis: the lateral shift of the microphone is part of the model. Seen from the driver, every position has a direction in the plane and a distance in 3D,

$$\varphi = \operatorname{atan2}(M_x - S_x,\ M_y - S_y) \ \text{(unwrapped to within } \pi \text{ of } \theta), \qquad \rho = |\mathbf{M} - \mathbf{S}|. \tag{18}$$

For the setup of Figure 5(b): $\delta = 145$ mm, $\ell = +120$ mm, $o = -205$ mm for the woofer and 0 for the midrange and tweeter, $h = -16.5$, -65 and $+65$ mm, $r_0 = 1000$ or 2000 mm.

4.2 Point-source re-propagation

Around each driver the response is assumed to be that of a point source with a directivity,

$$h(\varphi, \rho, t) = \frac{g(\varphi,\ t - \rho/c)}{\rho}, \tag{19}$$

so a measurement taken at (φ_k, ρ_k) is moved to the target distance ρ^* by a delay and a gain:

$$h_{k\to *}(t) = \frac{\rho_k}{\rho^*}\, h_k\!\left(t - \frac{\rho^* - \rho_k}{c}\right). \tag{20}$$

4.3 Angular interpolation

The measurements, sorted by φ_k, serve as supports (for a full circle, copies at $\varphi_k \pm 2\pi$ close the loop). For the target direction φ^* the two bracketing supports a, b, with the weights

$$u_a = 1 - u, \quad u_b = u, \qquad u = \frac{\varphi^* - \varphi_a}{\varphi_b - \varphi_a}, \tag{21}$$

are combined *after* both have been moved to ρ^*, so that they are in time. Their powers are interpolated, with the phase of their complex interpolation:

$$|H^*(f)|^2 = \sum_k u_k \, |H_{k\to *}(f)|^2, \qquad \arg H^*(f) = \arg \sum_k u_k \, H_{k\to *}(f). \tag{22}$$

Where the two supports are in phase, this is close to the linear interpolation of the complex responses. Where the angular step is too coarse for the pattern — interference ripples finer than the step, behind the cabinet at high frequency — they are not, and their complex mean cancels in part, digging notches that are in neither measurement; the mean of their powers does not (Section 7.4). The price is paid where the pattern is well sampled and changes level quickly with angle (a beaming driver's main lobe): there the mean of the powers reads slightly high (Section 6.2).

Outside the measured range the nearest support is used. The 0° measurement is the reference and is never changed. A measurement copied by symmetry at $-\theta$ stands for the mirror image of the one at θ and is placed at $(-\varphi_k, \rho_k)$ of that one. When the turntable reports the angle it really stopped at, that angle places the support; the target stays the nominal angle.

4.4 At another distance

The targets can lie on a larger circle than the one measured on, $r^* > r_0$: the drivers' directions and paths are then those seen from that circle, and the result is brought back to the level and the time of flight of the measurement distance, so that the reference point keeps its own measurements as they are:

$$g_k = \frac{\rho_k}{\rho^*}\,\frac{r^*}{r_0}, \qquad \tau_k = \frac{\rho^* - \rho_k - (r^* - r_0)}{c}. \tag{23}$$

Only the differences between drivers and directions change with the distance. A driver out of the plane ($h \neq 0$) is then seen from another angle across the plane than it was measured at ($\arctan h/r_0$ against $\arctan h/r^*$). The other plane's measurements give the change of its response between the two angles: the driver's power in any direction $\mathbf{d}$ is taken as its two planes' at the same angle ϑ off its axis,

$$P(\mathbf{d}) = (1-s)\,P_{\text{own}}(\pm\vartheta) + s\,P_{\text{other}}(\pm\vartheta), \tag{24}$$

where s is the squared sine of the angle by which $\mathbf{d}$ turns around the axis from the plane, and P_{own}, P_{other} are the two orbits' powers brought to 1 m, smoothed over $\pm 1/12$ octave and interpolated in angle. The response interpolated in the plane is multiplied by $\sqrt{P(\mathbf{d}^*)/P(\mathbf{d}_{\text{int}})}$, limited to ± 6 dB, from the direction the supports stand for to the target's. With one orbit measured, the driver is taken as symmetric about its axis (the other plane's powers are this plane's).

4.5 Implementation

The fractional delays are applied as linear phases. With $\tau_k = (\rho^* - \rho_k) f_s/c$ samples, $p = \lceil \max|\tau_k| \rceil + 1$ samples of margin and an FFT length $N_c = 2^{\lceil \log_2(L+2p+64) \rceil}$ (L the longest impulse response),

$$H_{k\to *}[m] = g_k \, X_k[m] \, e^{-\mathrm{j}2\pi \tilde{m}(\tau_k + p)/N_c}, \qquad \tilde{m} = \begin{cases} m & m \le N_c/2 \\ m - N_c & m > N_c/2 \end{cases}, \tag{25}$$

with $g_k = \rho_k/\rho^*$ and X_k the DFT of the zero-padded measurement; H^* follows from Eq. (22) bin by bin (the spectrum stays conjugate-symmetric) and the corrected impulse response is $\operatorname{Re}\operatorname{IDFT}\{H^*\}$ from sample p on.

4.6 Assumptions

The method assumes a point source on the baffle plane whose far-field directivity does not depend on distance. The cabinet edges violate it: they radiate from elsewhere than the driver, and how much of them the microphone hears changes with its distance. Directions are taken in the plane of rotation; the drivers' heights only enter the distances (and, at another distance, the change of angle across the plane).

5 Simulation protocol

Measurements. Horizontal orbit from $-180°$ to $180°$ in $10°$ steps (36 positions; $\pm 180°$ is one position), microphone at $d = 1\,\mathrm{m}$ and $d = 2\,\mathrm{m}$ from the baffle, at the reference height. For every position, Eq. (9) gives the complex spectrum of each driver on a 16 384-point grid at 48 kHz (2.93 Hz resolution); the impulse responses are its inverse real FFT (341 ms, far longer than any delay or decay).

Cases.

- **rigorous**: axis on the reference point — the reference;
- **uncorrected**: cabinet centered, microphone moved onto the reference axis;
- **corrected**: the same data after the correction with the true parameters;
- **direct sound only**: the three cases again with $P_{\mathrm{edge}} = 0$, to separate the correction's own errors from those due to diffraction;
- **variants**: corrected with wrong parameters (lateral offset omitted or with the wrong sign, woofer offset omitted, depth or lateral offset off by $\pm 10\,\mathrm{mm}$, microphone distance off by $+20\,\mathrm{mm}$, the drivers' heights ignored), with the complex responses interpolated instead of the powers, and with a $5°$ orbit;
- **at another distance**: the 1 m measurements rebuilt as seen from the 2 m circle, from the centered setup and from the rigorous one, compared with the rigorous 2 m data.

Metrics. Power spectra are smoothed to 1/24 octave,

$$P_X(\theta, f) = \left\langle |p_X(\theta, f')|^2 \right\rangle_{f' \in [f\,2^{-1/48},\, f\,2^{1/48}]}, \tag{26}$$

and compared with the rigorous case through the level error

$$\Delta L_X(\theta, f) = 10 \log_{10} \frac{P_X(\theta, f)}{P_{\mathrm{rigorous}}(\theta, f)}\ \mathrm{dB}. \tag{27}$$

Statistics (RMS, 95th percentile, maximum of $|\Delta L|$) are taken over the points that matter: in the driver's band (woofer 50 Hz–1 kHz, midrange 300 Hz–6 kHz, tweeter 2–20 kHz, summed loudspeaker 50 Hz–20 kHz), and no more than 20 dB below the rigorous on-axis response at the same frequency (this leaves out the deep nulls of a beaming cone, where a fraction of a degree changes the level by tens of dB).

CTA-2034-A curves from the horizontal orbit. Only the horizontal orbit is measured, so the standard's curves [2] are estimated from it (all averages on squared pressures):

$$\mathrm{LW}^2 = \tfrac{1}{7} \textstyle\sum_{\theta \in \{0, \pm 10, \pm 20, \pm 30\}} |p(\theta)|^2, \tag{28}$$

$$\mathrm{ER}^2 = \tfrac{1}{3}(\mathrm{FW}^2 + \mathrm{SW}^2 + \mathrm{RW}^2), \tag{29}$$

$$\mathrm{SP}^2 = \frac{\sum_\theta w(\theta)\, |p(\theta)|^2}{\sum_\theta w(\theta)}, \tag{30}$$

$$\mathrm{PIR}^2 = 0.12\,\mathrm{LW}^2 + 0.44\,\mathrm{ER}^2 + 0.44\,\mathrm{SP}^2, \qquad \mathrm{DI} = 10 \log_{10} \frac{\mathrm{LW}^2}{\mathrm{SP}^2}, \tag{31}$$

where the front-wall (FW), side-wall (SW) and rear-wall (RW) groups are themselves RMS averages over the angles 0, ± 10, ± 20, $\pm 30°$ (FW), ± 40 to $\pm 80°$ (SW) and ± 90, $180°$ (RW). The standard adds the $\pm 10°$ vertical directions to the listening window and the floor and ceiling bounces (vertical

Table 3: Sound-power weights of the horizontal orbit (CTA-2034-A Appendix C), normalized over the 36 directions; $w(180° - \theta) = w(\theta)$.

0°	0.0012	10°	0.0095	20°	0.0180	30°	0.0248	40°	0.0301
50°	0.0338	60°	0.0364	70°	0.0381	80°	0.0390	90°	0.0394

orbit) to the early reflections; using the horizontal weights alone for the sound power amounts to assuming the vertical orbit equal to the horizontal one. The sound-power weights are those of the standard's Appendix C,

$$w(\theta) = \Omega(\theta + 5°) - \Omega(\theta - 5°), \qquad \Omega(\alpha) = 4C - 2\pi, \tag{32}$$

with $\Omega(\alpha)$ the solid angle of the spherical square bounded by the great circles at $\pm\alpha$ horizontally and vertically (C its corner angle), symmetric about 90° and doubled at 90° (Table 3). These estimates are the same for every case, so their differences measure the measurement geometry only.

6 Results

6.1 Uncorrected: what the centered setup does to the data

Figure 8 (center column) and Figure 9 show the uncorrected data against the rigorous ones. Off axis the responses are too low — by the distance ratio, up to 2.5 dB behind the cabinet at 1 m — and, where the response changes quickly with angle (beaming of the woofer and of the midrange's horn-loaded dome, diffraction ripple behind the cabinet), they show features of the wrong direction. Within the relevant points the summed loudspeaker is off by 2.45 dB RMS at 1 m and 1.24 dB at 2 m (Table 4).

6.2 Corrected, direct sound only: the method itself

Without edge diffraction every driver is exactly a point source with a directivity, as the correction assumes. What remains is the interpolation between measured directions: 0.10 dB RMS on the summed loudspeaker at 1 m, 0.08, 0.23 and 0.13 dB for the woofer, midrange and tweeter (Table 4, "direct only"). Most of it is the price of interpolating powers on a well-sampled pattern: where the main lobe of a beaming driver falls by several dB from one support to the next, the mean of their powers reads higher than the response between them. Interpolating the complex responses instead leaves 0.03 dB on the sum and 0.08 dB on the midrange; Section 7.4 shows what that choice costs with real diffraction. The drivers' heights (±65 mm for the domes) do not matter at 1 m and beyond: ignoring them changes the RMS figures by less than 0.01 dB. In this idealized world the correction is, for practical purposes, exact.

6.3 Corrected, with diffraction: the open question

With the cabinet edges, the residual grows: 0.42 dB RMS for the tweeter and 0.24 dB for the summed loudspeaker at 1 m, with local deviations up to 2.9 dB (Table 4, Figures 10 and 11). Figure 12 shows where: in the front half ($|\theta| \le 90°$) no angle exceeds 0.25 dB RMS on the summed loudspeaker (0.55 dB on the beaming midrange, the cost of the power interpolation seen above); the residual is behind the cabinet ($|\theta| \ge 120°$), at mid and high frequencies, where the edge waves dominate the response — up to 1.2 dB RMS at one angle for the tweeter at 1 m, 2.9 dB at 2 m. The woofer, whose band is low, is hardly affected.

Two mechanisms are at work. First, the edges are not at the driver: the correction moves the edge waves by the driver's change of distance, not by their own, so their timing relative to the direct sound keeps the error of the measured position — visible in the late part of the impulse responses (Figure 13). This error shrinks with distance. Second, behind the cabinet the response

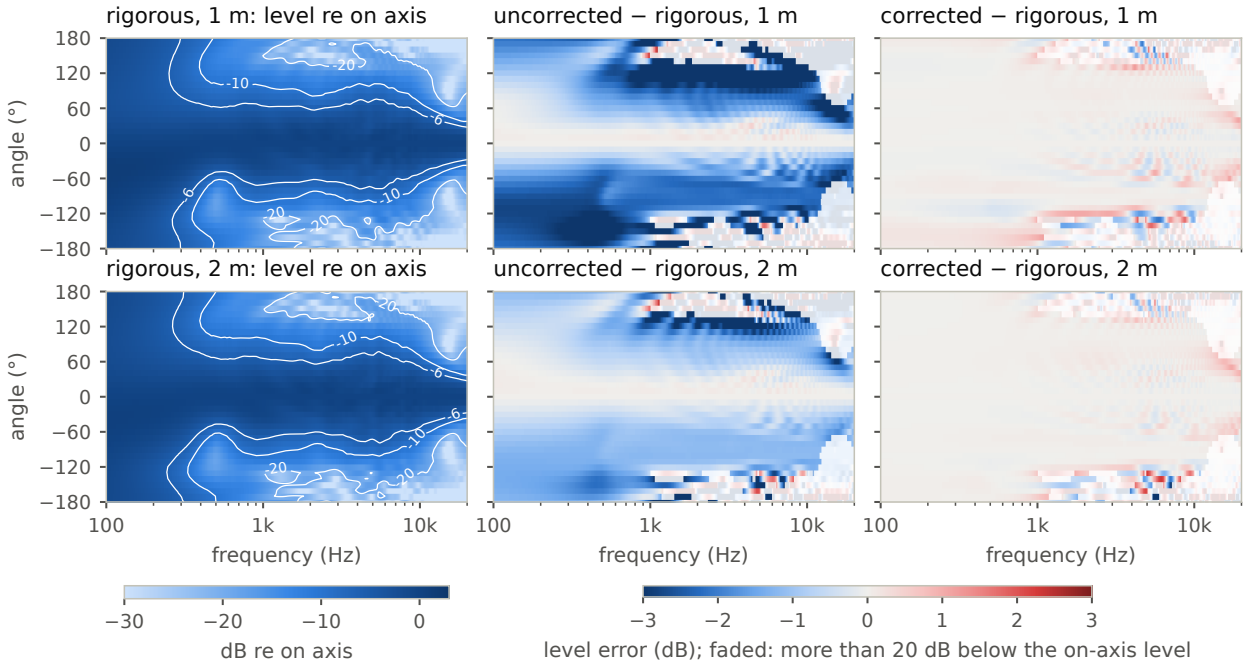


Figure 8: Summed loudspeaker. Left: rigorous horizontal polar map, normalized to the on-axis response (contours at −6, −10, −20 dB). Center and right: level error of the uncorrected and corrected data; points more than 20 dB below the on-axis level are faded.

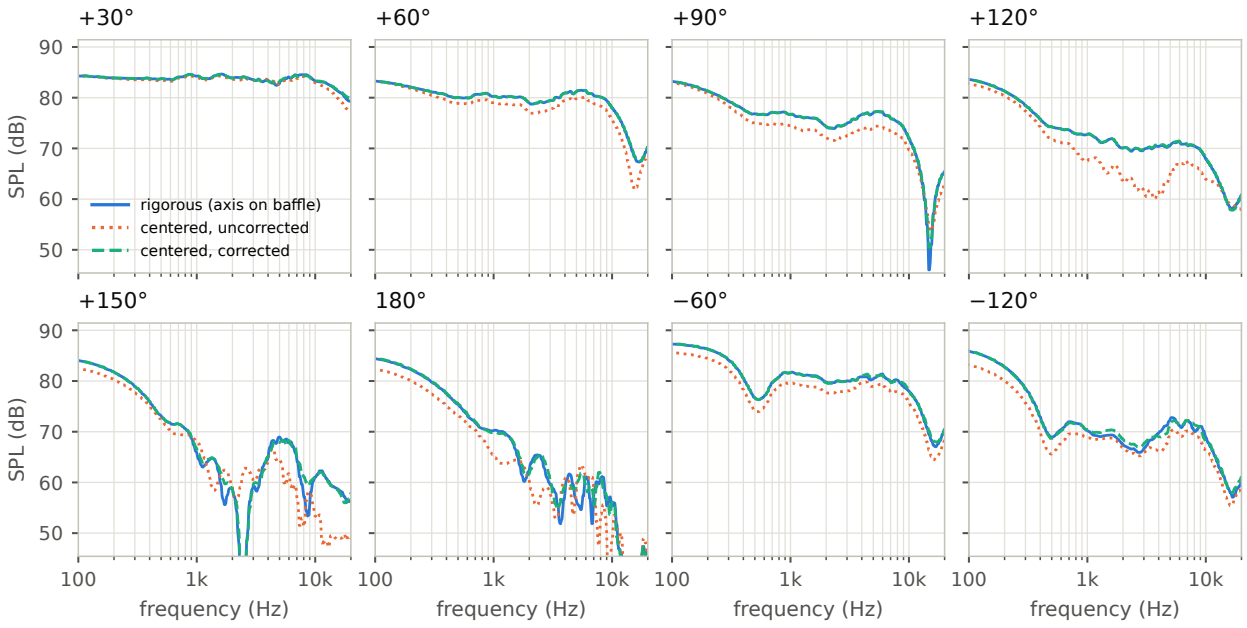


Figure 9: Summed loudspeaker at selected angles, 1 m, 1/12-octave smoothing.

Table 4: Level error $|\Delta L|$ against the rigorous measurement over the relevant points (dB): RMS, 95th percentile and maximum.

			uncorrected			corrected		
d	model	driver	RMS	p95	max	RMS	p95	max
1 m	DED	woofer	1.84	2.93	8.7	0.09	0.19	0.8
1 m	DED	midrange	2.39	4.27	20.1	0.29	0.57	2.1
1 m	DED	tweeter	3.30	6.79	22.8	0.42	0.99	2.9
1 m	DED	3-way sum	2.45	4.07	25.6	0.24	0.49	2.2
1 m	direct only	woofer	2.95	7.10	9.1	0.08	0.24	0.3
1 m	direct only	midrange	2.76	6.16	9.2	0.23	0.30	2.2
1 m	direct only	tweeter	2.72	5.93	9.2	0.13	0.28	0.8
1 m	direct only	3-way sum	2.77	6.30	9.3	0.10	0.22	0.8
2 m	DED	woofer	0.98	1.62	7.7	0.04	0.07	0.5
2 m	DED	midrange	1.19	2.09	9.2	0.22	0.39	2.2
2 m	DED	tweeter	1.71	3.57	12.0	0.41	0.81	4.9
2 m	DED	3-way sum	1.24	2.07	9.1	0.22	0.33	4.8
2 m	direct only	woofer	1.36	3.16	4.0	0.05	0.11	0.2
2 m	direct only	midrange	1.36	2.96	4.5	0.20	0.25	2.0
2 m	direct only	tweeter	1.34	2.88	4.5	0.12	0.23	0.9
2 m	direct only	3-way sum	1.33	3.16	4.3	0.08	0.16	0.9

is an interference pattern of many edge waves, with lobes about 10° wide at 5 kHz (λ/W) and narrower above; the correction interpolates between measured directions 10° apart, which cannot follow it. This one does not shrink with distance: at 2 m the summed loudspeaker keeps 0.22 dB RMS and its worst local deviation even grows to 4.8 dB, in a narrow region near −170° around 5–6 kHz.

6.4 Angular resolution

Halving the step to 5° addresses the second mechanism. At 2 m it brings the summed loudspeaker from 0.22 to 0.09 dB RMS and its worst deviation from 4.8 to 1.3 dB; the tweeter goes from 0.41 to 0.17 dB RMS. At 1 m the gain is smaller (0.24 to 0.19 dB RMS), because there the first mechanism, which no angular step can fix, takes a larger share.

6.5 Spatial averages and Predicted In-Room Response

The CTA-2034-A curves average many directions and the residual errors, concentrated behind the cabinet and changing sign from one frequency to the next, largely cancel (Figure 14, Table 5). Uncorrected, the centered setup lowers the sound power by up to 1.4 dB at 1 m (every off-axis microphone position is farther away), the PIR by up to 1.02 dB and shifts the directivity index by 1.2 dB; at 2 m these errors halve. Corrected, the PIR stays within 0.12 dB (1 m) and 0.08 dB (2 m) of the rigorous one, the sound power within 0.18 dB and the directivity index within 0.15 dB. Of those 0.18 dB on the sound power, 0.15 come from the power interpolation's bias on the beaming drivers (they remain with the direct sound alone); interpolating the complex responses gives 0.14 dB with the edges, and the PIR within 0.14 dB.

6.6 Wrong parameters and the laterally shifted microphone

The correction needs the geometry it is given (Figure 15, Table 6). Errors of ±10 mm on the depth or the lateral offset, or of 20 mm on the microphone distance, change the RMS errors by a few hundredths of a dB and the PIR by less than 0.1 dB: measuring with a tape is good enough.

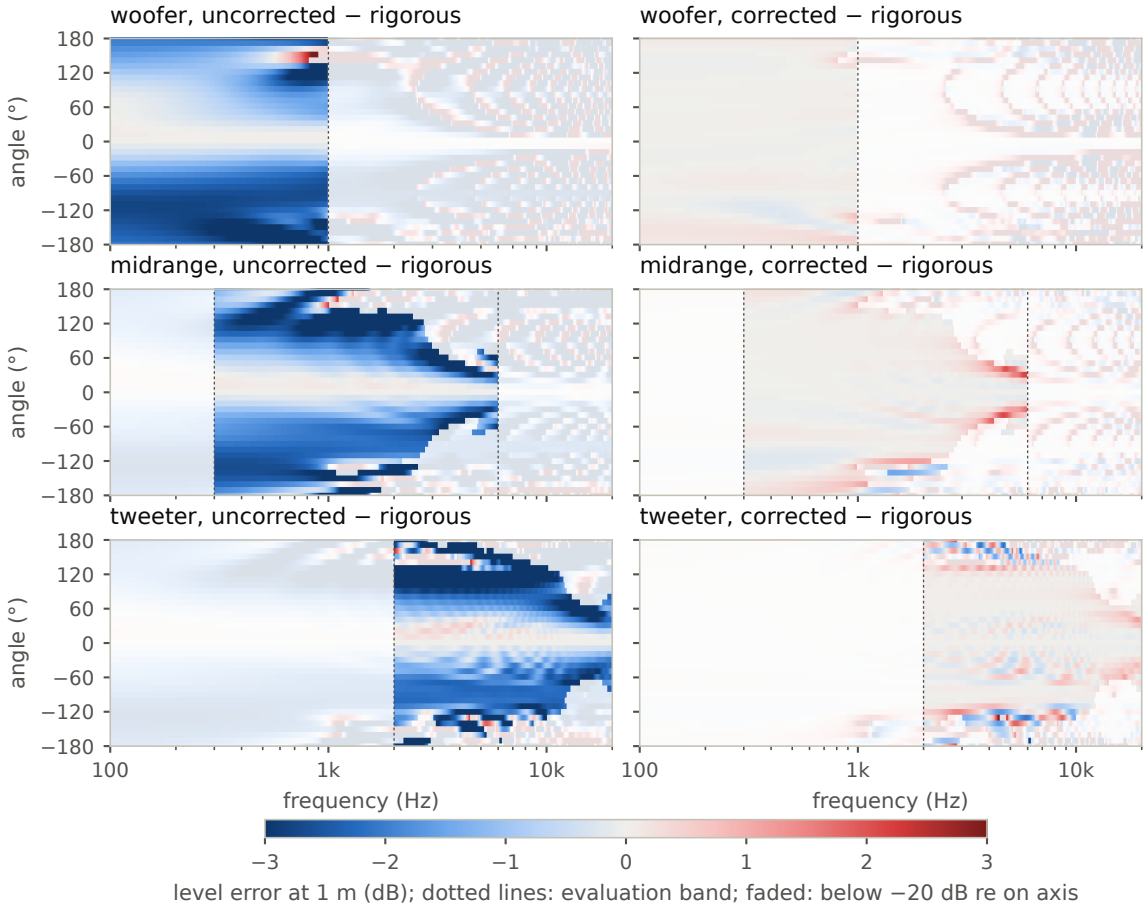


Figure 10: Level error of each driver at 1 m, uncorrected (left) and corrected (right). Dotted lines: evaluation band. Faded: more than 20 dB below the on-axis response.

Table 5: Largest deviation from the rigorous case, 100 Hz–20 kHz, of the horizontal-orbit CTA-2034-A curves of the summed loudspeaker (dB).

d	model	data	LW	ER	SP	PIR	DI
1 m	DED	uncorrected	0.51	0.95	1.39	1.02	1.19
1 m	DED	corrected	0.17	0.11	0.18	0.12	0.15
1 m	direct only	uncorrected	0.51	0.87	1.33	0.86	1.11
1 m	direct only	corrected	0.02	0.04	0.15	0.05	0.13
2 m	DED	uncorrected	0.26	0.44	0.66	0.48	0.58
2 m	DED	corrected	0.08	0.08	0.14	0.08	0.12
2 m	direct only	uncorrected	0.26	0.40	0.63	0.40	0.53
2 m	direct only	corrected	0.01	0.03	0.11	0.04	0.10

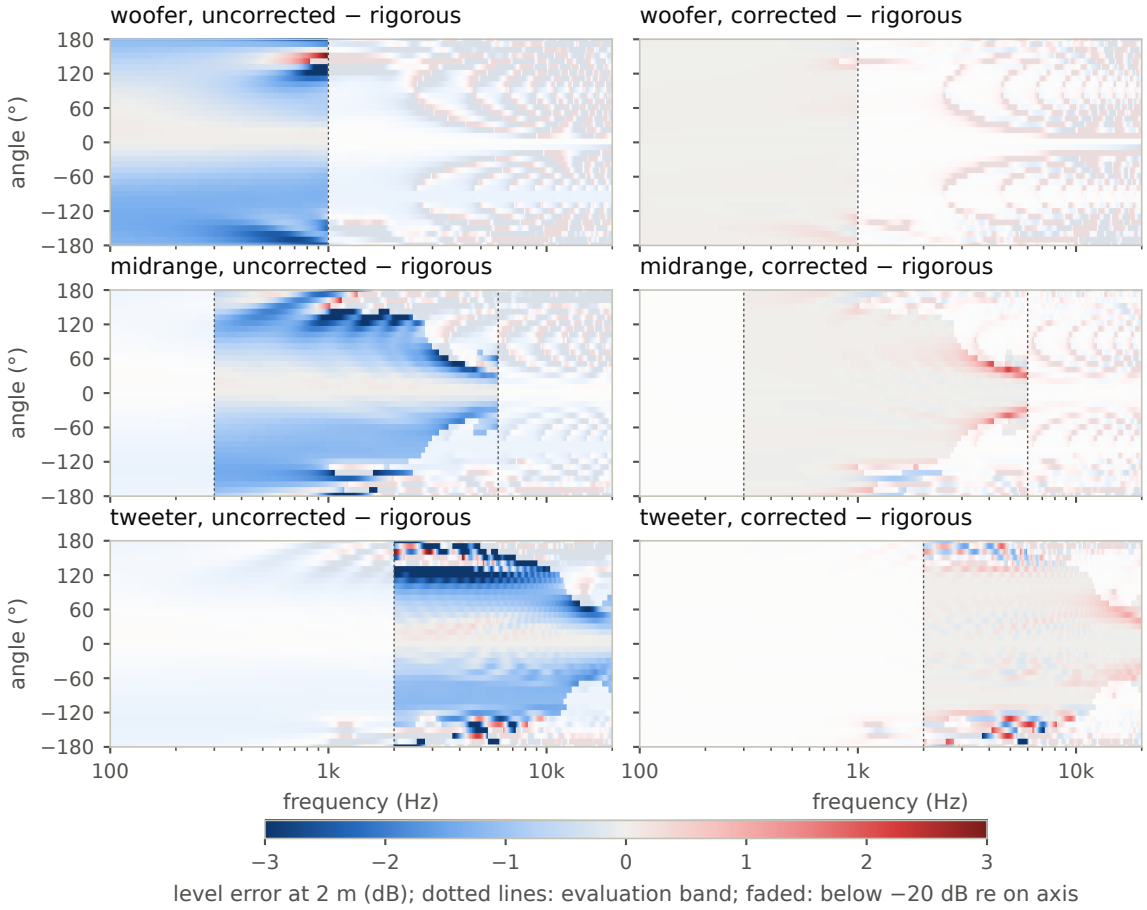


Figure 11: Same as Figure 10 at 2 m.

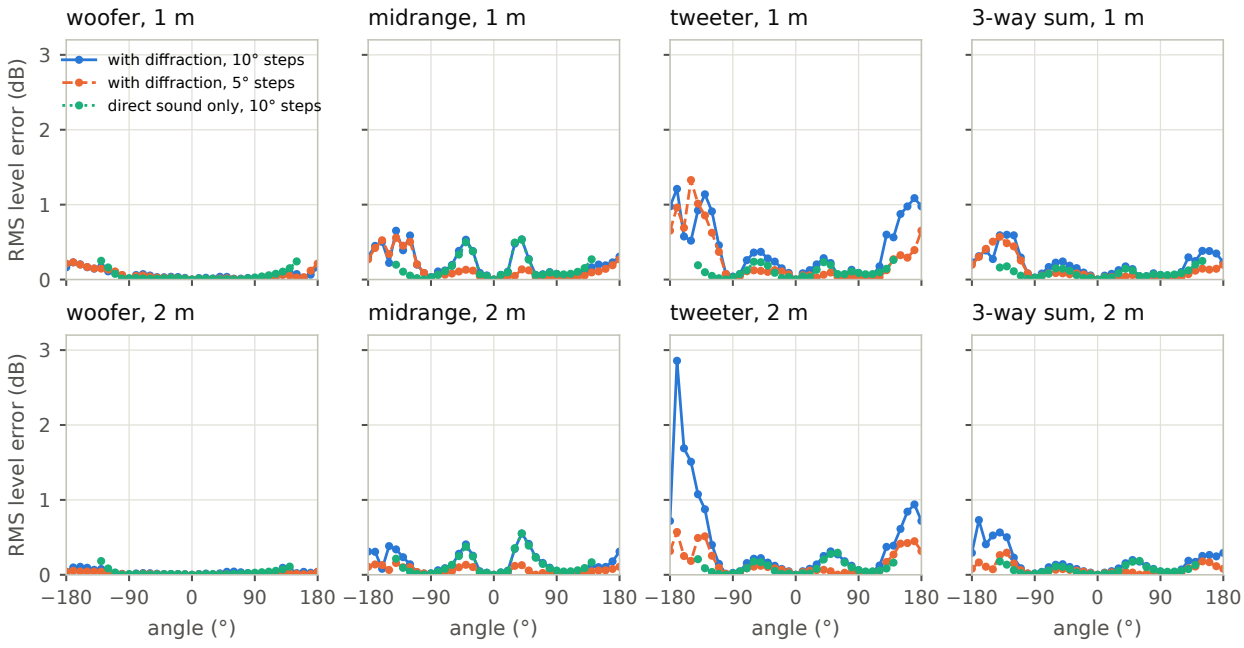


Figure 12: RMS level error of the corrected data at each angle (over the relevant frequencies), with the DED diffraction and 10° or 5° steps, and for the direct sound alone.

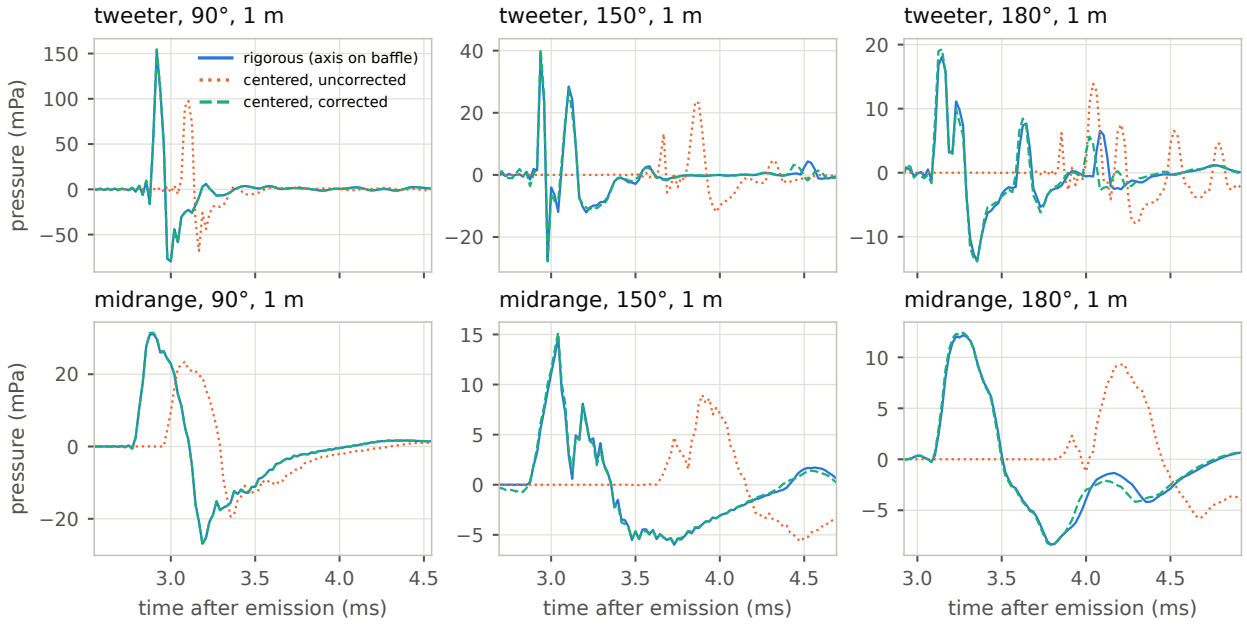


Figure 13: Impulse responses at 1 m (time from emission). The uncorrected responses arrive late and low; the corrected ones line up with the rigorous ones, except for the later edge-wave arrivals behind the cabinet.

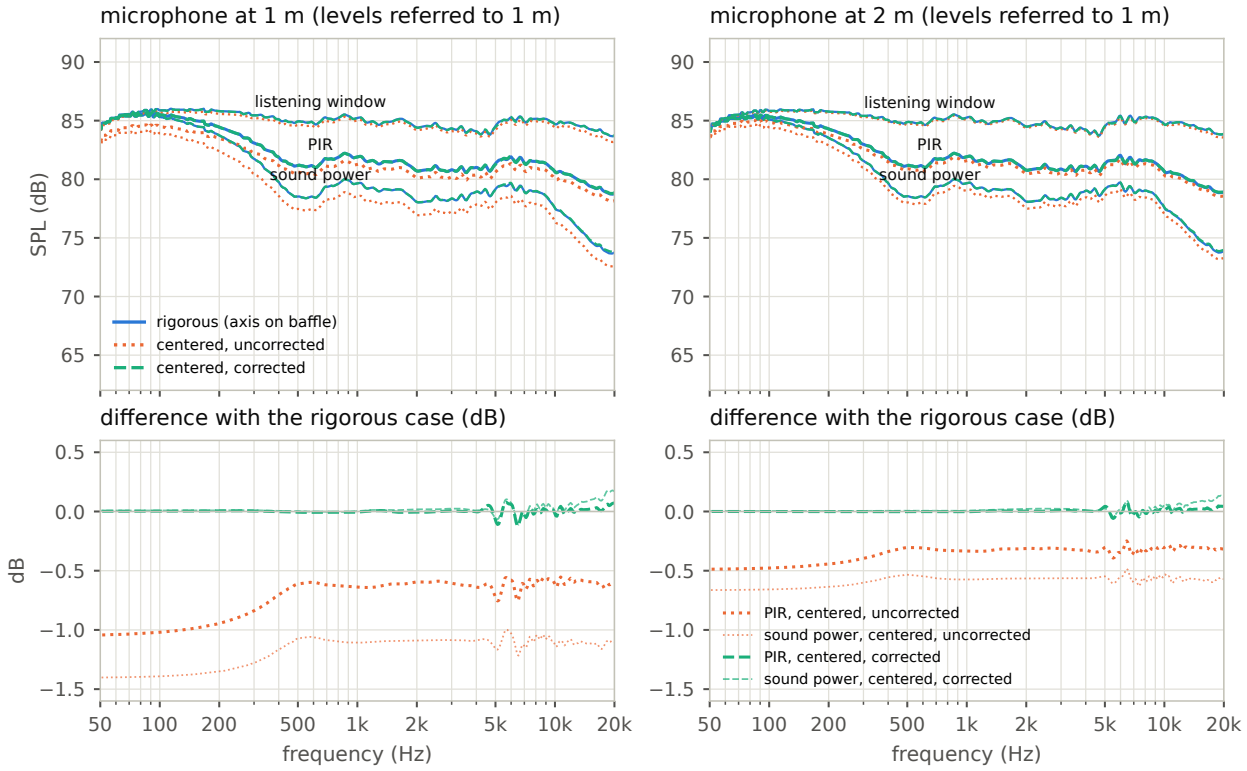


Figure 14: Top: listening window, PIR and sound power of the summed loudspeaker from the horizontal orbit (levels referred to 1 m). The corrected curves lie on the rigorous ones. Bottom: differences with the rigorous case.

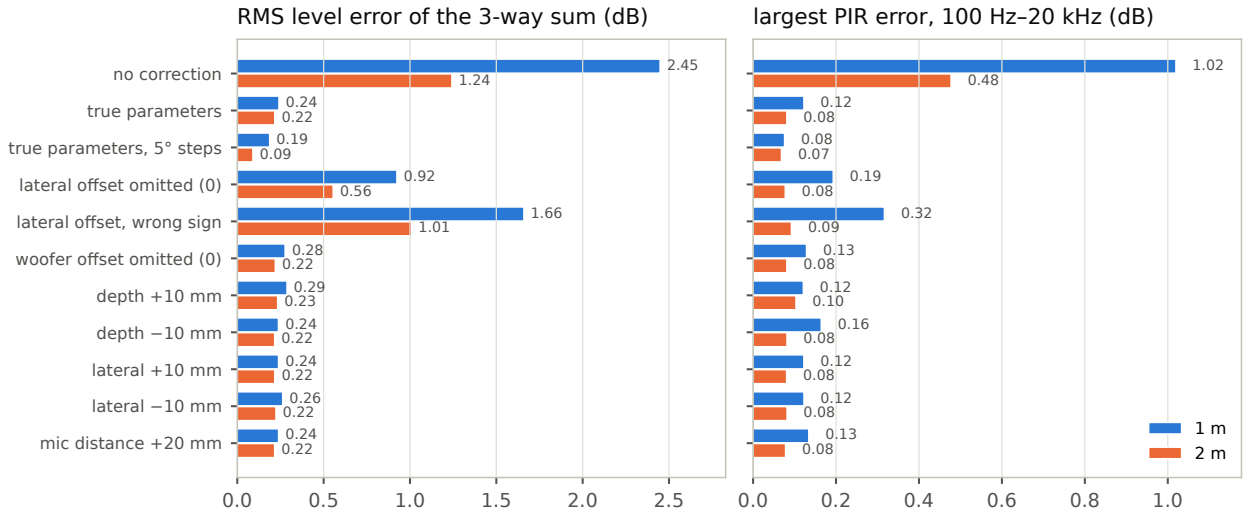


Figure 15: Effect of the correction parameters: RMS level error of the summed loudspeaker (left) and largest PIR deviation (right), at 1 m and 2 m.

The lateral offset itself is another matter. Because the microphone has been moved sideways to face the reference point, the model must know that the reference axis is $\ell = 120$ mm from the rotation axis, on the + side. Omitting it ($\ell = 0$, i.e. correcting only for the depth, as for a symmetric loudspeaker) places the measurements up to 11° and 110 mm away from where they were taken, at 1 m, and leaves the summed loudspeaker 0.92 dB RMS off, four times the residual with the right offset (0.24 dB); entering it with the wrong sign doubles the error of placement (21°, 210 mm) and gives 1.66 dB RMS (the tweeter 2.33 dB against 0.42). The spatial averages hardly notice: the PIR stays within 0.19 and 0.32 dB, against 0.12 dB. With the complex responses interpolated, the same mistakes were worse than no correction at all (the PIR 3.8 and 3.3 dB off at 1 m, against 1.02 dB uncorrected): the supports, moved with the wrong delays, no longer lined up in time and their complex mean cancelled. Interpolated as powers they do not cancel, and what remains is the wrong direction and level. The sign convention matters: ℓ is positive when the reference axis lies on the side the microphone faces at +90°. With the convention of this study (positive angles toward the tweeter side) the reference axis is on the + side; a turntable that counts angles the other way needs $\ell = -120$ mm and a woofer offset of +205 mm.

The woofer's own offset (−205 mm) matters less: omitting it raises the woofer's error from 0.09 to 0.23 dB RMS at 1 m, and the summed loudspeaker hardly notices, the woofer being crossed over at 450 Hz where its directivity varies slowly. Ignoring the drivers' heights changes nothing measurable.

6.7 Rebuilt at 2 m from measurements at 1 m

CTA-2034-A measures at 2 m. Seen from there, a driver beside the reference axis is closer to the nominal angles than seen from 1 m: the woofer, 205 mm aside, is 11.6° off its axis when the microphone is on the reference axis at 1 m, 5.9° at 2 m. Rebuilding the 1 m measurements as from the 2 m circle (Section 4.4, the driver taken as symmetric about its axis for the change of angle across the plane, since only the horizontal orbit is measured; leaving that change out makes no difference here) fixes that geometry. Table 7 compares the results with the rigorous 2 m data (levels brought back to 1 m). The woofer's error falls from 0.74 dB RMS (the rigorous 1 m data used as they are) to 0.24 dB (the same data rebuilt) and 0.16 dB (the centered 1 m data rebuilt); the summed loudspeaker from 0.58 to 0.26 dB, not far from the 0.22 dB of the 2 m measurements corrected at 2 m. The PIR, on the other hand, stays 0.18 dB off, against 0.08 dB measured at 2 m: the cabinet's diffraction, heard from 1 m, is not the one heard from 2 m, and no point-source model moves it from one distance to the other. Section 7.5 shows the same with the boundary-element

Table 6: RMS level error per driver and for the summed loudspeaker, and largest PIR deviation, for the correction variants (dB).

	1 m					2 m				
	woofer	mid	tweeter	sum	PIR	woofer	mid	tweeter	sum	PIR
no correction	1.84	2.39	3.30	2.45	1.02	0.98	1.19	1.71	1.24	0.48
true parameters	0.09	0.29	0.42	0.24	0.12	0.04	0.22	0.41	0.22	0.08
true parameters, 5° steps	0.09	0.19	0.29	0.19	0.08	0.02	0.07	0.17	0.09	0.07
lateral offset omitted (0)	0.59	0.93	1.41	0.92	0.19	0.38	0.54	0.84	0.56	0.08
lateral offset, wrong sign	1.14	1.39	2.33	1.66	0.32	0.72	0.96	1.44	1.01	0.09
woofer offset omitted (0)	0.23	0.29	0.42	0.28	0.13	0.07	0.22	0.41	0.22	0.08
depth +10 mm	0.15	0.35	0.46	0.29	0.12	0.08	0.25	0.42	0.23	0.10
depth −10 mm	0.11	0.26	0.41	0.24	0.16	0.06	0.21	0.40	0.22	0.08
lateral +10 mm	0.12	0.27	0.41	0.24	0.12	0.06	0.22	0.41	0.22	0.08
lateral −10 mm	0.09	0.32	0.45	0.26	0.12	0.05	0.23	0.41	0.22	0.08
mic distance +20 mm	0.08	0.28	0.41	0.24	0.13	0.04	0.22	0.41	0.22	0.08
complex responses interpolated	0.09	0.21	0.41	0.24	0.14	0.03	0.14	0.41	0.23	0.07
drivers' heights ignored	0.09	0.29	0.42	0.24	0.12	0.04	0.22	0.41	0.22	0.08

Table 7: Results at 2 m against the rigorous 2 m data: RMS level error per driver and for the summed loudspeaker, and largest PIR deviation (dB), horizontal orbit, 10° steps.

	woofer	mid	tweeter	sum	PIR
rigorous at 1 m, used as they are	0.74	0.35	0.61	0.58	0.19
rigorous at 1 m, rebuilt at 2 m	0.24	0.35	0.62	0.40	0.21
centered at 1 m, rebuilt at 2 m	0.16	0.29	0.41	0.26	0.18
... without the change of angle across the plane	0.16	0.29	0.41	0.26	0.17
centered at 2 m, corrected	0.04	0.22	0.41	0.22	0.08

solution, more clearly. The centered data come out closer to the 2 m truth than the rigorous ones: behind the cabinet, where diffraction depends most on the distance, their microphone was farther away.

7 A boundary-element check

Two parts of the model are approximations, and both matter most where the residual of the correction lives: the drivers radiate as points with a far-field directivity, and the DED model represents the cabinet by dipoles along the front edges only, which describes the field behind the cabinet least well. This section repeats the study on a variant of the loudspeaker computed with the boundary-element method (BEM), which solves the radiation of the whole cabinet — every face and edge, the drivers as extended rigid surfaces — with neither approximation, from 20 Hz to 16 kHz.

7.1 A two-way variant

The BEM model is a two-way in the same cabinet (Figure 16): the SIG225-4 woofer at $x = -85$ mm and the SICA tweeter at $x = +120$ mm, both at mid-height ($y = 0$). Dropping the midrange removes a source and, with both drivers on the cabinet's horizontal middle plane, makes the problem symmetric about the plane of the horizontal orbit, which halves it. The reference point is the tweeter center and the microphone is at its height: both drivers lie in the plane of rotation. The correction's parameters are those of the three-way: depth 145 mm, lateral offset +120 mm, woofer offset −205 mm, tweeter offset 0.

The radiators are rigid and move along their axis: the woofer is a recessed cone of radius

$a = \sqrt{S_d/\pi} = 83.8\,\text{mm}$, 35 mm deep at $r = 30\,\text{mm}$ and 40 mm at the center (the dust cap), the tweeter the dome of Section 2.3 (28 mm, 7.5 mm high), both rims flush with the baffle.

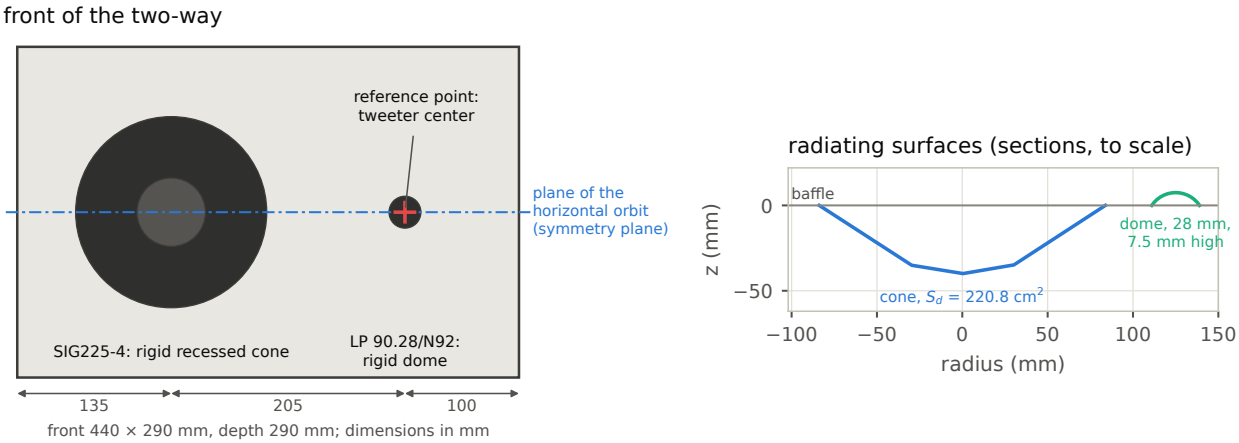


Figure 16: The two-way variant solved by the BEM: same cabinet, woofer and tweeter on its horizontal middle plane, which is the plane of the horizontal orbit and a plane of symmetry.

Drivers. For each driver moving with a volume velocity of $1\,\text{m}^3/\text{s}$, the BEM gives the complex pressure $Z_i(\mathbf{M}, f)$ at any point $\mathbf{M}$ (Pa per m^3/s). The volume velocity comes from the datasheet response of Section 2.2 (the woofer in its 24 L box):

$$p_i(\mathbf{M}, f) = U_i(f)\, Z_i(\mathbf{M}, f), \qquad U_i(f) = \frac{H_i(f)}{q_i(f)}, \qquad q_i(f) = \frac{\mathrm{j}\omega\rho_0}{2\pi r_{\text{ref}}}\, A_i(0, k), \tag{33}$$

where q_i is the on-axis far-field pressure of the same radiator in an infinite baffle per unit volume velocity, according to the aperture model of Eq. (7) (with the cone's profile $z(r) < 0$ for the woofer). Were that model exact, p_i would be $H_i\, D_i(\theta, k)\, (r_{\text{ref}}/r)\, e^{-\mathrm{j}kr}$ in an infinite baffle, the direct term of Eq. (9): the BEM and the DED model of this two-way share the same drivers and differ only by how they radiate and diffract. The DED model is evaluated with the cone's profile too.

Crossover. Fourth-order Linkwitz–Riley at 2 kHz, the woofer high-passed at 40 Hz, the filters designed as in Section 2.5 from the BEM's on-axis responses at 1 m: the target times the minimum-phase inverse of each driver's 1/6-octave smoothed magnitude. The same filters serve both models.

7.2 The boundary-element solution

The BEM data were computed with Huygens, a boundary-element solver under development whose code is not public; the pressures at every microphone position are in the repository (`data/bem/two_way.npz`), with the scripts that turn them into the results below. The method:

- exterior Neumann problem of the Helmholtz equation around the closed cabinet, rigid, each driver's surface moving along its axis, the other one at rest (one right-hand side per driver);
- Burton–Miller formulation [7] (the direct integral equation combined with its normal derivative, with a coupling of modulus $1/k$), so that no frequency falls on an interior resonance of the box; Galerkin discretization, pressure piecewise linear and normal derivative piecewise constant on flat triangles, hypersingular operator in Maue's weak form, singular integrals by the Sauter–Schwab transformations [5];
- uniform mesh of the whole surface, edges of $\lambda/6$ at the top of each band of frequencies (bands of a ratio 1.26 in edge length), 50 mm at most; the half model and its mirror image;

- assembly with the near and singular pairs of triangles in double precision on the processor and the distant pairs in single precision on the GPU; dense LU factorization in double precision.

The 464 frequencies run from 20 Hz to 16 kHz in 1/48-octave steps. At 16 kHz the surface has 162 898 triangles and the half model 40 937 unknowns (30.0 GB, 606 s per frequency). The microphone positions are those of the study, extended: the horizontal and vertical orbits of both setups, at 1, 2 and 4 m, every 2.5° (1728 points). In the centered setup the vertical orbit is measured with the cabinet on its side on the turntable: its axis runs through the cabinet's center, 145 mm behind the baffle, level with the drivers. The sweep took 3.8 h on a Mac mini (Apple M4 Pro, 64 GB), 62 % of it in the top band. A first sweep at 432 of these points gave the same pressures, to the $6 \cdot 10^{-8}$ of their storage in single precision.

Two checks on the data. At 20 Hz, on the 2 m circle, each driver radiates the pressure of a monopole of the same volume velocity, $\omega\rho_0 U/(4\pi r)$, to within ±7 %, a variation explained below. Where the mesh changes (by 1.26 in edge length) the levels at the relevant points of the horizontal orbits move by 0.1 to 0.5 dB RMS, about as much as from one frequency to the next within a band: the mesh is fine enough for the comparisons that follow.

All the results of this section are computed at the BEM frequencies for both models, and the power spectra are smoothed to 1/24 octave as before (on this grid, the mean of three neighboring frequencies). The evaluation bands are 50 Hz–4 kHz for the woofer, 1–16 kHz for the tweeter and 50 Hz–16 kHz for the sum.

7.3 What the DED model misses

The low-frequency acoustic center lies ahead of the baffle. At 50 Hz the BEM level at 0° exceeds the level at 180° by 2.3 dB at 1 m and 1.2 dB at 2 m: a monopole about 130 mm in front of the baffle, at both distances — the low-frequency acoustic center described by Vanderkooy [6]. The DED model puts it about 40 to 60 mm ahead. It is the variation of the 20 Hz check above.

Baffle step. On the reference axis the BEM woofer rises 4.5 dB above its half-space level around 1 kHz, the DED model 2 dB (Figure 17).

The dome radiates wider. The rigid dome of the BEM is much wider than the aperture model of Eq. (7): on the tweeter side of the orbit its response falls by 6 dB at 95° at 6.35 kHz (DED: 75°), at 80° at 10 kHz (55°) and at 50° at 16 kHz (35°), and its on-axis level is 2 to 3 dB lower above 6 kHz for the same volume velocity (Figures 17 and 19). The aperture model lets each ring of the cap radiate as if it lay in a plane baffle; the protruding cap radiates sideways. With the drivers' volume velocities taken from that model, the BEM tweeter therefore sits 2 to 3 dB below its datasheet on axis above 6 kHz; the crossover's equalization absorbs it.

Stronger diffraction. More sound reaches the edges, and the BEM's polar patterns carry interference ripples that the DED model hardly shows. On the woofer side of the front quadrant (−70° to −20°) the ripple has a period of about 25° at 3 kHz and 10° at 6.35 kHz, with ±1 to 3 dB: a secondary source some 330 mm from the tweeter, the far edge of the baffle (Figure 19). Behind the cabinet the patterns are finer still. Directivity aside (levels referred to 0°, Figure 18, Table 8), the two models differ by 3.36 dB RMS in front of the tweeter and 5.29 dB behind it at 1 m.

7.4 The correction on the BEM data

Table 10 and Figures 20 and 21 give the residuals; Table 11 the spatial averages.

The same two-way without its cabinet (point sources in an infinite baffle, the DED model's direct term) is corrected to within 0.09 dB RMS on the sum at 1 m with 10° steps, as the three-way's direct sound: what follows is the cabinet's doing.

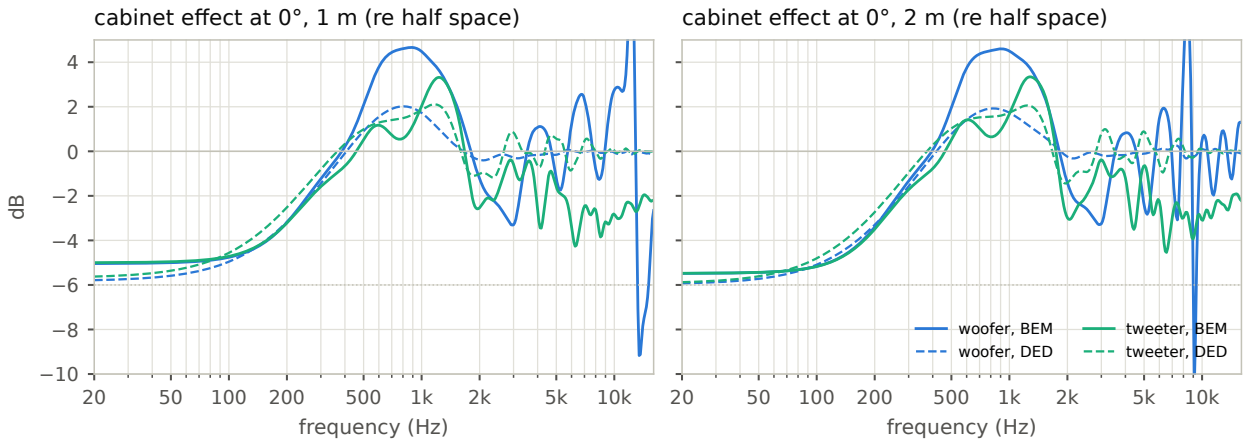


Figure 17: Each driver at 0° (the tweeter's axis), BEM and DED, referred to the same driver in an infinite baffle according to the aperture model: the cabinet effect, plus for the BEM the difference between the rigid dome and its aperture model above a few kHz. At low frequency the BEM's acoustic center, 130 mm ahead of the baffle, raises the level at 0°.

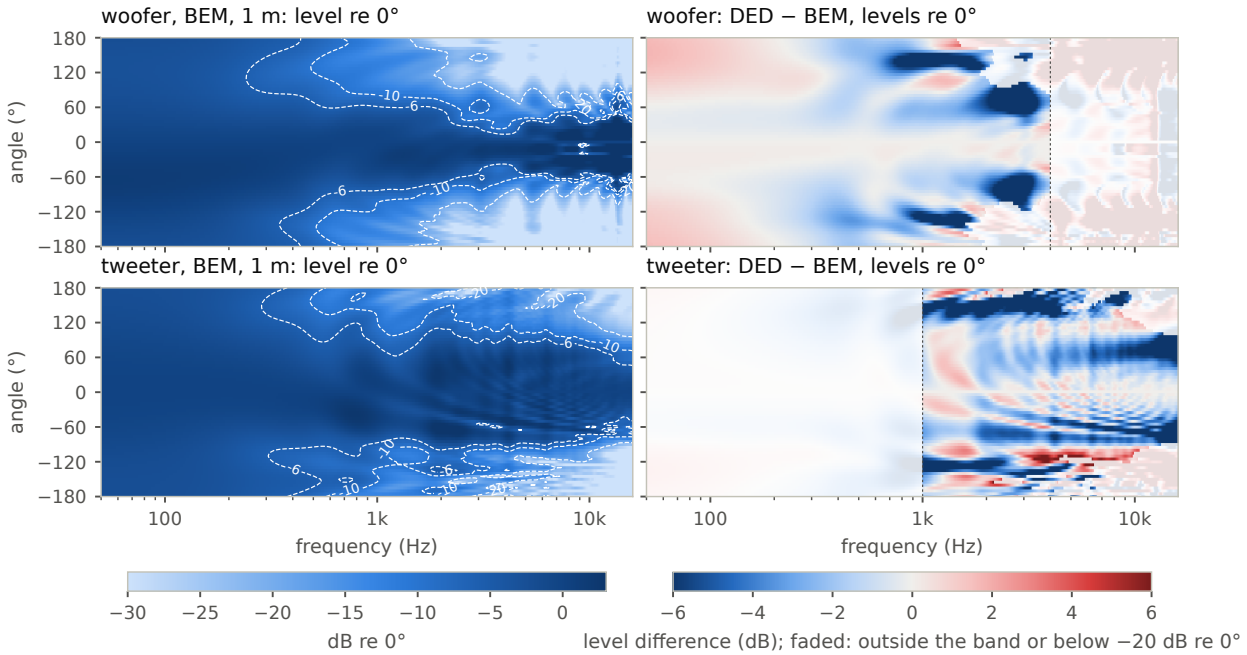


Figure 18: Left: BEM polar maps of the two drivers at 1 m (5° steps), levels re 0°. Right: the DED model's directivity minus the BEM's.

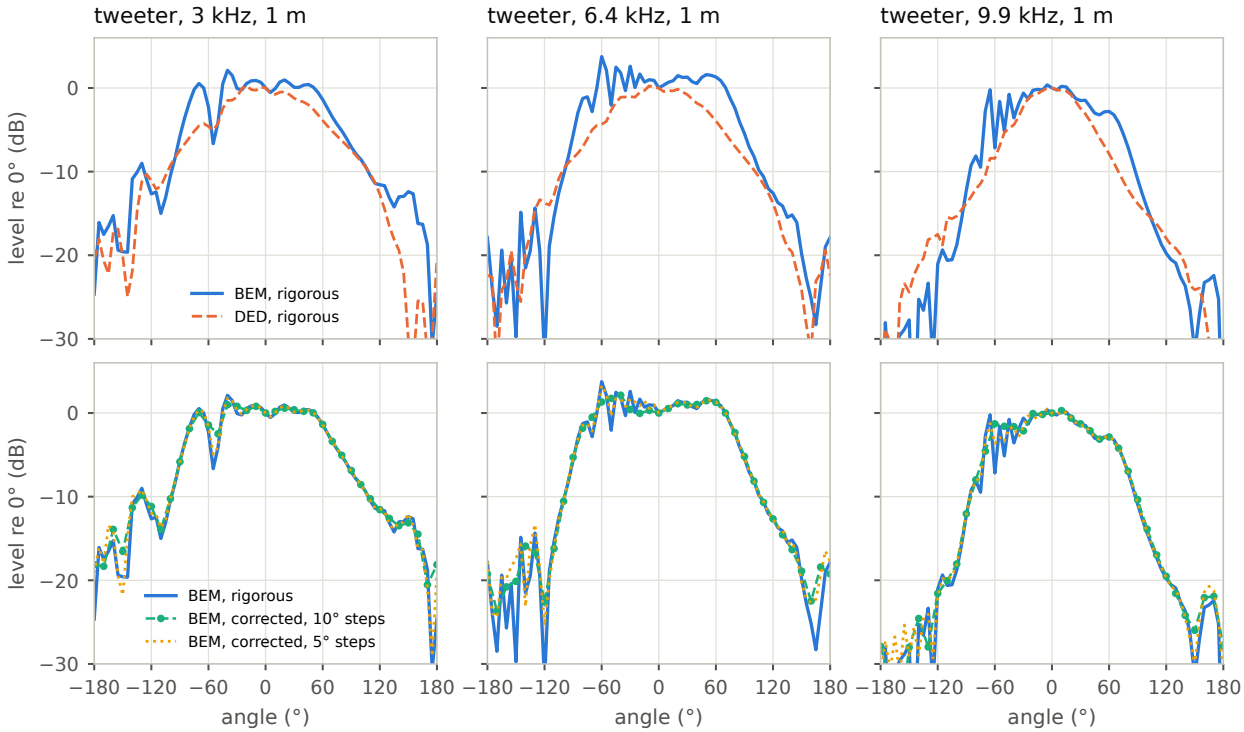


Figure 19: Tweeter at 1 m, level re 0° against angle at 3, 6.35 and 10 kHz. Top: BEM and DED. Bottom: what the correction rebuilds from the centered measurements of the BEM, with 10° and 5° steps. Negative angles are on the woofer's side.

Table 8: DED against BEM, rigorous setup, levels referred to 0°: RMS and 95th percentile of the difference over the relevant points, in front of the baffle ($|\theta| \leq 90°$) and behind it (dB).

		front		behind	
d		RMS	P95	RMS	P95
1 m	woofer	2.45	4.99	3.01	6.18
1 m	tweeter	3.36	6.31	5.29	11.85
1 m	2-way sum	2.55	5.23	4.07	9.58
2 m	woofer	2.48	5.26	3.24	6.71
2 m	tweeter	3.33	6.12	5.61	12.12
2 m	2-way sum	2.49	5.17	4.25	9.95
4 m	woofer	2.50	5.40	3.33	7.11
4 m	tweeter	3.32	6.12	5.70	12.21
4 m	2-way sum	2.47	5.17	4.32	10.04

The woofer is corrected about as well as in the DED model of this two-way: 0.26 dB RMS at 1 m, 0.15 dB at 2 m. The low-frequency acoustic center ahead of the baffle does not get in the way: the spatial averages of the corrected data stay within 0.03 dB of the rigorous ones up to 1 kHz.

The tweeter and the summed loudspeaker keep more than in the DED model: 0.96 and 0.62 dB RMS at 1 m (DED: 0.57 and 0.35 dB), 0.99 and 0.60 dB at 2 m. The residual sits where the BEM's patterns are finest: on the woofer's side, in front and behind, and directly behind the cabinet (Figure 21). Three mechanisms add up:

- *Angular sampling.* 5° steps bring the tweeter to 0.63 dB RMS at 1 m and 0.43 dB at 2 m, 2.5° steps to 0.58 and 0.22 dB; the sum goes to 0.41 and 0.28, then 0.38 and 0.14 dB (Figure 23, Table 14). With 10° steps the residual is the same at 1, 2 and 4 m (0.62, 0.60 and 0.58 dB on the sum): the interference patterns have the same angular period at every distance. At 4 m it is half the uncorrected error (1.20 dB): on individual curves, interpolating across 10° then costs about as much as the geometry gains.
- *Powers or complex responses.* Where the two measurements that bracket a direction are no longer coherent — the rear patterns at 10° steps — their complex mean cancels in part, down to −20 dB notches (Figure 19, −130° around 6 kHz). The interpolated powers (Eq. 22) avoid it: interpolating the complex responses instead leaves 1.81 dB RMS on the tweeter at 1 m with 10° steps, against 0.96 dB, and 1.21 against 0.62 dB on the sum (Table 10). With 2.5° steps the neighbors are coherent again and both interpolations agree; in the DED model it makes no difference, and on the point sources the complex interpolation is slightly better (0.03 dB on the sum).
- *Secondary sources.* The far edge, 330 mm from the tweeter, re-radiates the sound that grazes the baffle; the correction moves that wave as if it came from the tweeter, so the ripple it makes in front lands at the wrong angle. This part remains with fine steps: with 2.5° steps the sum keeps 0.38 dB RMS at 1 m. It shrinks with distance: 0.14 dB at 2 m, 0.09 dB at 4 m.

Spatial averages. Up to 3 kHz the corrected CTA-2034-A curves remain within about 0.1 dB of the rigorous ones, as in the DED model. Between 4 and 13 kHz the residual concentrates in narrow bands: the PIR deviates by up to 0.55 dB at 1 m and 0.32 dB at 2 m, the listening window by 0.82 and 0.44 dB (Figure 22, Table 11); uncorrected, the PIR is 1.24 and 0.65 dB off. With 5° steps the PIR comes within 0.29 dB at 1 m and 0.28 dB at 2 m, with 2.5° steps within 0.15, 0.10 and 0.07 dB at 1, 2 and 4 m. Part of those tenths belongs to the standard's own sampling: the far edge's ripple has a period close to 10° in front at 6–8 kHz, and the rigorous listening window computed from its seven 10° directions differs by up to 0.6–0.7 dB at 7–8 kHz from the same window averaged over every 2.5°.

The vertical orbit measured with the cabinet on its side needs the depth only: the drivers are level with the axis, and the woofer is 205 mm beside the orbit's plane, which the distances in 3D take into account (Table 15). Uncorrected, the sum is 1.86 dB RMS off at 1 m; corrected, 0.38 dB with 10° steps and 0.26 dB with 5° (DED: 0.19 and 0.12 dB): the top and bottom edges, 145 mm from the drivers, play the part of the far edge. The woofer's offset from the plane costs little: 0.22 dB RMS corrected, 0.25 dB when it is taken as in the plane.

Both orbits, as CTA-2034-A measures them, both on the turntable and both corrected: the PIR is within 0.34 dB at 1 m, 0.17 dB at 2 m and 0.09 dB at 4 m with 10° steps, 0.18, 0.16 and 0.10 dB with 5° steps, 0.08, 0.06 and 0.05 dB with 2.5°; uncorrected, 0.95, 0.47 and 0.25 dB (Table 12). The DED model of the same two-way puts the corrected PIR within 0.05 dB at 1 m.

Parameters. The conclusions of Section 6.6 stand (Table 13): an error of ±10 mm on the depth or on the lateral offset, or of 20 mm on the distance, changes the PIR by less than 0.1 dB; omitting the lateral offset, or giving it the wrong sign, leaves the sum 1.39 or 1.97 dB RMS off at 1 m, two to three times the residual, while the PIR hardly moves (0.52 and 0.50 dB, against 0.55). Crossed over at 2 kHz instead of 450 Hz, the woofer needs its own offset a little more than in the three-way: without it, 0.30 dB RMS instead of 0.26.

The whole loudspeaker. Measured at once, through its crossover, and corrected as one source at the reference point (the tweeter), the two-way keeps 0.68 dB RMS on the horizontal orbit at 1 m with 10° steps, against 0.62 dB with each driver measured and corrected on its own, and 0.41 against 0.38 dB with 2.5° steps; the PIR from both orbits is within 0.33 dB (0.34 dB). The difference sits around the crossover: 0.85 against 0.70 dB RMS between 1 and 4 kHz. Corrected at once, the woofer and the tweeter keep the relative delay of the position the microphone was at; with the powers interpolated, that error cancels nothing. It should grow with the drivers' distance from the reference point, relative to the microphone distance (not simulated here). The drivers measured on their own remain the cleaner way, and the one a crossover is designed from.

7.5 At another distance

The data hold the same orbits at 1, 2 and 4 m, so the results rebuilt at a larger distance than measured (Section 4.4) can be checked against the rigorous data at that distance, each orbit with the other's measurements for the angles across its plane. Table 9 gives the RMS error of the sum on both orbits and the largest deviation of the PIR from both orbits, with 10° and 2.5° steps.

Rebuilt at 2 m, the centered 1 m data come out about twice as close to the rigorous 2 m data as the rigorous 1 m data themselves: 0.65 dB RMS on the sum of the horizontal orbit against 1.34 dB, the PIR within 0.37 against 0.57 dB. But measured at 2 m and corrected there, with 2.5° steps, the sum is within 0.14 dB and the PIR within 0.06 dB, where the 1 m data rebuilt keep 0.67 and 0.35 dB whatever the step. What the rebuild cannot move is the cabinet's near field:

- at low frequency, the acoustic center about 130 mm ahead of the baffle: going from 1 to 2 m, the level falls by $20 \log_{10} \frac{2-0.13}{1-0.13} = 6.6$ dB in front and 5.5 dB behind, not 6.0 dB; the rebuilt woofer is ±0.3 dB RMS off below 1 kHz, in front and behind;
- at high frequency, the diffraction pattern, whose rear lobes and far-edge ripple differ between 1 and 2 m: between 4 and 16 kHz the rebuilt sum keeps 0.8 dB RMS in front and 1.6 dB behind.

Without the cabinet (point sources, Table 9 does not show them), the rebuild is exact to within 0.1 dB. For the vertical orbit, where the woofer is 205 mm beside the plane and seen at 11.6° across it from 1 m but 5.9° from 2 m, the horizontal orbit's measurements give that change of angle. They change nothing measurable here: the sum is 0.52 dB RMS off with them, 0.50 dB without. The woofer's directivity changes little over 6° in its band, and the factor, read from measurements that carry the cabinet's diffraction, removes about as much as it adds. The same holds for the three-way's domes, ±65 mm from the plane (Table 7). The correction matters for drivers farther from the plane and closer to the microphone: a woofer 250 mm under the tweeter is seen 14° off the plane at 1 m and 7° at 2 m.

The two-way's spinorama. For reference, Figure 24 shows the CTA-2034-A curves of the BEM two-way from both orbits at 1 m.

7.6 What changes

The BEM confirms what the correction does below about 3 kHz, where the woofer and the crossover region are: the spatial averages of the corrected data match the rigorous ones within 0.1 dB. Above,

Table 9: The BEM two-way at a larger distance than measured, against the rigorous data at that distance (levels brought back to the measurement distance): RMS level error of the sum, horizontal (H) and vertical (V) orbits, and largest PIR deviation from both orbits, 100 Hz–16 kHz (dB). The last row of each group is measured at the larger distance and corrected there.

	10° steps			2.5° steps		
	sum, H	sum, V	PIR	sum, H	sum, V	PIR
measured at 1 m, results at 2 m						
rigorous, used as measured	1.34	0.92	0.57	1.39	0.85	0.57
rigorous, rebuilt	1.17	0.93	0.41	1.22	0.85	0.41
centered, rebuilt	0.65	0.52	0.37	0.67	0.49	0.35
centered at that distance, corrected	0.60	0.30	0.17	0.14	0.10	0.06
measured at 1 m, results at 4 m						
rigorous, used as measured	1.94	1.32	0.82	1.99	1.29	0.82
rigorous, rebuilt	1.69	1.38	0.44	1.73	1.34	0.44
centered, rebuilt	1.04	0.87	0.56	1.22	0.94	0.44
centered at that distance, corrected	0.58	0.18	0.09	0.09	0.05	0.05
measured at 2 m, results at 4 m						
rigorous, used as measured	0.75	0.45	0.31	0.72	0.44	0.31
rigorous, rebuilt	0.68	0.47	0.32	0.64	0.45	0.32
centered, rebuilt	0.71	0.36	0.24	0.45	0.34	0.28
centered at that distance, corrected	0.58	0.18	0.09	0.09	0.05	0.05

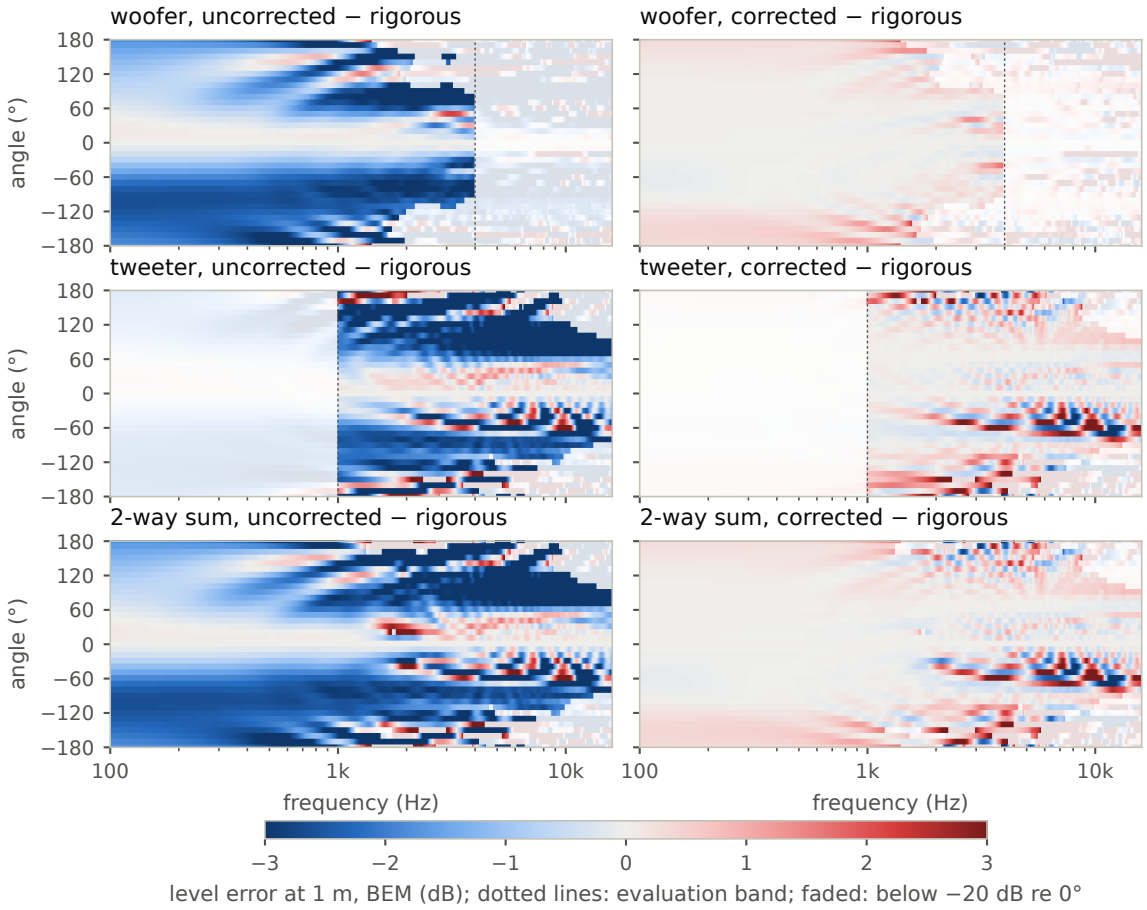


Figure 20: Level error of the BEM data at 1 m, uncorrected (left) and corrected (right), 10° steps.

Table 10: Level error of the two-way, BEM, DED and point sources (the DED model's direct term), horizontal orbit at 1, 2 and 4 m: uncorrected, corrected (10° steps), corrected with the complex responses interpolated instead of the powers (10°), corrected with 5° steps (dB, over the relevant points).

			uncorrected			corrected			complex		5° steps	
d	model		RMS	P95	max	RMS	P95	max	RMS	max	RMS	max
1 m	BEM	woofer	2.02	3.20	19.9	0.26	0.49	2.3	0.25	3.2	0.22	2.1
1 m	BEM	tweeter	3.24	6.20	25.5	0.96	2.28	6.0	1.81	22.7	0.63	5.2
1 m	BEM	2-way sum	2.56	4.61	26.0	0.62	1.27	6.2	1.21	22.7	0.41	5.7
1 m	DED	woofer	2.17	3.63	13.0	0.15	0.30	1.1	0.11	0.9	0.10	0.7
1 m	DED	tweeter	3.38	6.91	26.9	0.57	1.29	5.4	0.56	4.9	0.43	4.7
1 m	DED	2-way sum	2.66	4.67	26.7	0.35	0.62	5.5	0.36	4.9	0.26	4.7
1 m	point sources	woofer	2.80	5.95	9.1	0.14	0.26	1.2	0.05	0.4	0.03	0.2
1 m	point sources	tweeter	2.84	6.80	9.3	0.10	0.25	0.7	0.04	0.3	0.02	0.2
1 m	point sources	2-way sum	2.89	6.70	16.8	0.09	0.22	0.7	0.03	0.3	0.02	0.2
2 m	BEM	woofer	1.05	1.81	6.9	0.15	0.29	1.9	0.14	2.1	0.07	0.6
2 m	BEM	tweeter	2.51	4.03	34.8	0.99	1.88	12.6	1.59	22.1	0.43	3.0
2 m	BEM	2-way sum	1.79	2.58	34.9	0.60	0.92	12.6	1.05	21.7	0.28	3.0
2 m	DED	woofer	1.09	1.94	9.4	0.11	0.22	1.1	0.07	0.5	0.04	0.4
2 m	DED	tweeter	2.01	4.08	25.3	0.56	1.01	8.4	0.62	8.9	0.20	2.3
2 m	DED	2-way sum	1.45	2.49	25.6	0.35	0.38	8.4	0.41	9.0	0.12	2.0
2 m	point sources	woofer	1.31	3.09	4.0	0.10	0.19	0.8	0.03	0.3	0.03	0.2
2 m	point sources	tweeter	1.39	3.29	4.6	0.09	0.19	0.9	0.03	0.4	0.02	0.2
2 m	point sources	2-way sum	1.35	3.13	6.6	0.07	0.13	0.9	0.03	0.4	0.02	0.2
4 m	BEM	woofer	0.55	0.95	3.6	0.15	0.31	1.6	0.26	5.0	0.04	0.6
4 m	BEM	tweeter	1.74	2.43	21.6	0.94	1.76	10.3	1.52	18.7	0.42	5.9
4 m	BEM	2-way sum	1.20	1.44	22.4	0.58	0.81	10.3	1.02	19.0	0.27	5.9
4 m	DED	woofer	0.54	0.99	2.5	0.10	0.21	0.9	0.07	0.6	0.03	0.3
4 m	DED	tweeter	1.01	2.16	10.4	0.52	0.95	5.8	0.60	6.6	0.19	2.2
4 m	DED	2-way sum	0.72	1.28	9.0	0.31	0.29	5.5	0.36	6.0	0.11	2.2
4 m	point sources	woofer	0.66	1.60	2.1	0.09	0.21	0.7	0.03	0.3	0.03	0.2
4 m	point sources	tweeter	0.69	1.61	2.2	0.09	0.20	0.8	0.03	0.3	0.02	0.2
4 m	point sources	2-way sum	0.68	1.60	3.0	0.08	0.19	0.8	0.02	0.3	0.02	0.2

Table 11: Largest deviation from the rigorous case, 100 Hz–16 kHz, of the horizontal-orbit CTA-2034-A curves of the two-way (dB).

d	model	data	LW	ER	SP	PIR	DI
1 m	BEM	uncorrected	0.87	1.23	1.65	1.24	1.48
1 m	BEM	corrected	0.82	0.52	0.79	0.55	0.81
1 m	DED	uncorrected	0.57	0.95	1.40	1.02	1.20
1 m	DED	corrected	0.15	0.07	0.11	0.08	0.13
2 m	BEM	uncorrected	0.46	0.65	0.89	0.65	0.72
2 m	BEM	corrected	0.44	0.30	0.41	0.32	0.39
2 m	DED	uncorrected	0.29	0.44	0.66	0.48	0.60
2 m	DED	corrected	0.11	0.06	0.09	0.06	0.09
4 m	BEM	uncorrected	0.21	0.33	0.46	0.33	0.38
4 m	BEM	corrected	0.21	0.15	0.20	0.16	0.19
4 m	DED	uncorrected	0.14	0.21	0.32	0.23	0.31
4 m	DED	corrected	0.06	0.04	0.06	0.04	0.06

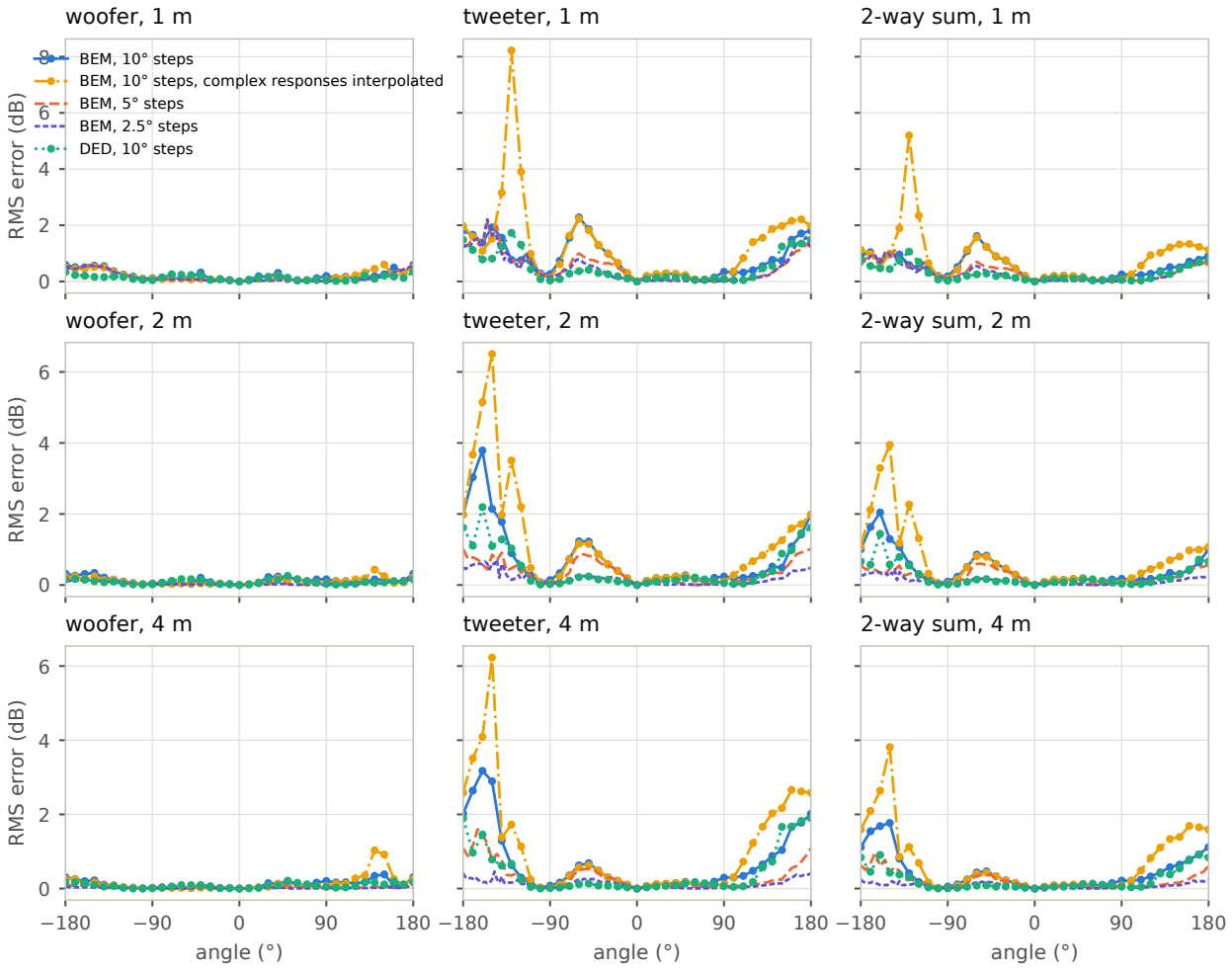


Figure 21: RMS level error of the corrected data at each angle, at 1, 2 and 4 m: BEM with 10° steps, the same with the complex responses interpolated instead of the powers, BEM with 5° and 2.5° steps, and the DED model of the same two-way. Negative angles are on the woofer's side.

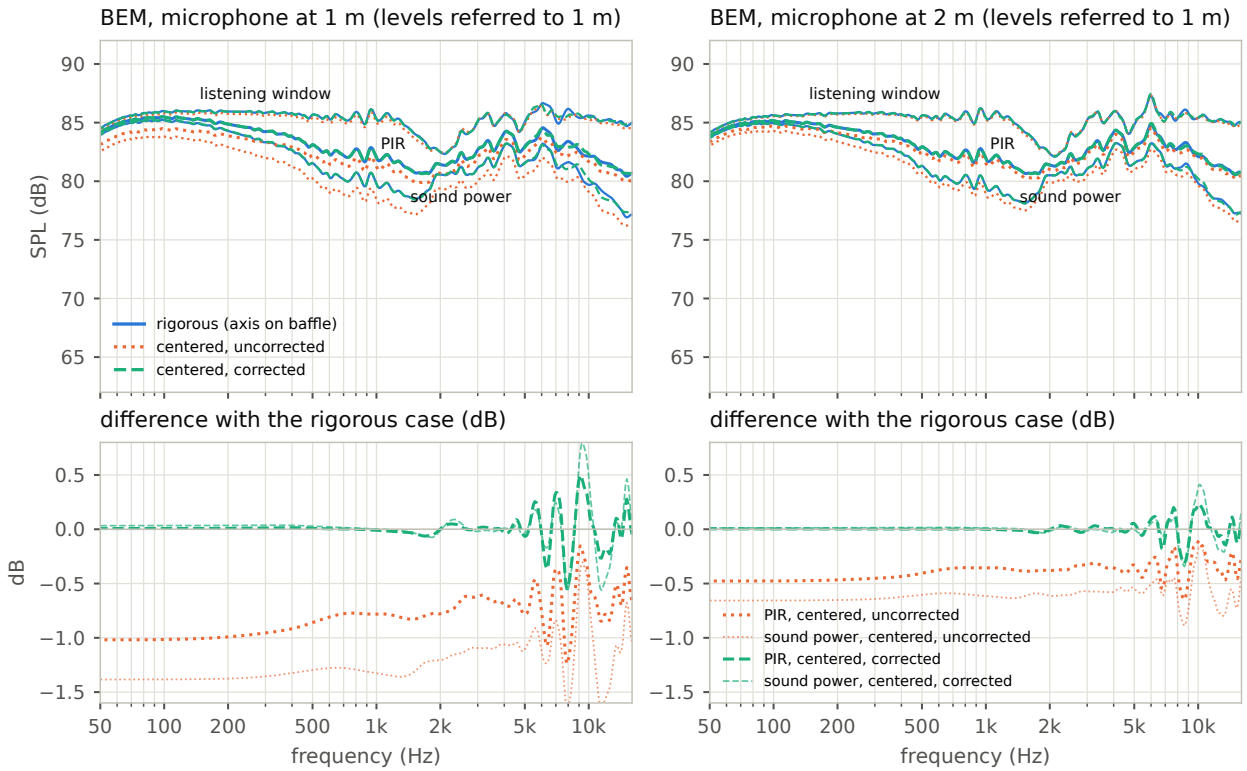


Figure 22: Listening window, PIR and sound power of the BEM two-way from the horizontal orbit, and their differences with the rigorous case.

Table 12: The same with both orbits, as CTA-2034-A measures them, both measured on the turntable (the vertical one with the cabinet on its side): uncorrected, corrected (10° steps), and corrected with 5° steps (dB).

d	model	horizontal	vertical	LW	ER	SP	PIR	SPDI	ERDI
1 m	BEM	uncorrected	uncorrected	0.66	0.93	1.37	0.95	1.33	0.79
1 m	BEM	corrected	corrected	0.66	0.31	0.40	0.34	0.57	0.47
1 m	BEM	corrected, 5°	corrected, 5°	0.42	0.15	0.16	0.18	0.33	0.30
1 m	DED	uncorrected	uncorrected	0.42	0.87	1.25	0.85	1.10	0.71
1 m	DED	corrected	corrected	0.12	0.08	0.12	0.05	0.14	0.10
1 m	DED	corrected, 5°	corrected, 5°	0.07	0.03	0.03	0.03	0.06	0.06
2 m	BEM	uncorrected	uncorrected	0.35	0.53	0.72	0.47	0.73	0.44
2 m	BEM	corrected	corrected	0.35	0.14	0.21	0.17	0.32	0.26
2 m	BEM	corrected, 5°	corrected, 5°	0.34	0.14	0.16	0.16	0.26	0.24
2 m	DED	uncorrected	uncorrected	0.20	0.45	0.63	0.42	0.58	0.39
2 m	DED	corrected	corrected	0.08	0.06	0.09	0.04	0.10	0.07
2 m	DED	corrected, 5°	corrected, 5°	0.07	0.04	0.04	0.04	0.06	0.04
4 m	BEM	uncorrected	uncorrected	0.16	0.27	0.37	0.25	0.36	0.22
4 m	BEM	corrected	corrected	0.17	0.07	0.10	0.09	0.16	0.12
4 m	BEM	corrected, 5°	corrected, 5°	0.18	0.08	0.09	0.10	0.13	0.13
4 m	DED	uncorrected	uncorrected	0.10	0.23	0.31	0.21	0.30	0.20
4 m	DED	corrected	corrected	0.05	0.03	0.06	0.03	0.06	0.04
4 m	DED	corrected, 5°	corrected, 5°	0.05	0.02	0.03	0.02	0.04	0.03

Table 13: RMS level error and largest PIR deviation of the BEM two-way for the correction variants (dB).

	1 m				2 m			
	woofer	tweeter	sum	PIR	woofer	tweeter	sum	PIR
no correction	2.02	3.24	2.56	1.24	1.05	2.51	1.79	0.65
true parameters	0.26	0.96	0.62	0.55	0.15	0.99	0.60	0.32
true parameters, 5° steps	0.22	0.63	0.41	0.29	0.07	0.43	0.28	0.28
complex interpolation	0.25	1.81	1.21	0.63	0.14	1.59	1.05	0.36
complex interpolation, 5° steps	0.21	0.70	0.47	0.36	0.06	0.62	0.41	0.34
true parameters, 2.5° steps	0.22	0.58	0.38	0.15	0.06	0.22	0.14	0.10
complex interpolation, 2.5° steps	0.21	0.58	0.37	0.15	0.05	0.22	0.14	0.11
lateral offset omitted (0)	0.68	1.94	1.39	0.52	0.45	1.67	1.12	0.29
lateral offset, wrong sign	1.28	2.69	1.97	0.50	0.78	1.93	1.44	0.28
woofer offset omitted (0)	0.30	0.96	0.65	0.54	0.16	0.99	0.60	0.31
depth +10 mm	0.33	1.01	0.66	0.54	0.18	1.00	0.61	0.29
depth −10 mm	0.22	0.94	0.61	0.60	0.14	0.98	0.59	0.34
lateral +10 mm	0.28	0.95	0.63	0.55	0.16	0.97	0.59	0.32
lateral −10 mm	0.25	1.00	0.64	0.55	0.15	1.01	0.61	0.31
mic distance +20 mm	0.24	0.96	0.62	0.56	0.15	0.99	0.60	0.32

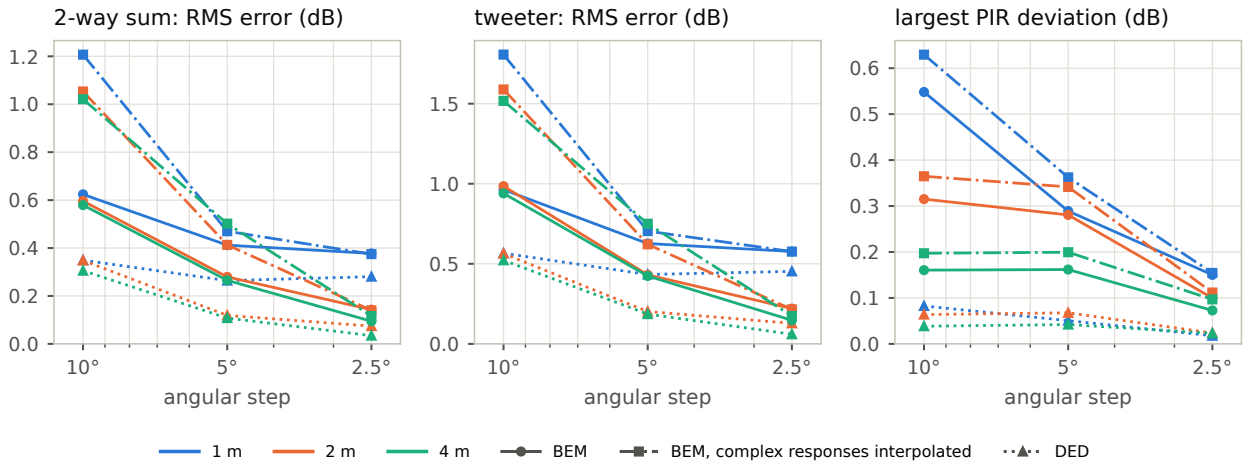


Figure 23: Residual of the corrected two-way against the angular step, at 1, 2 and 4 m: BEM, BEM with the complex responses interpolated instead of the powers, and the DED model. Horizontal orbit.

Table 14: Residual of the corrected two-way against the angular step, horizontal orbit: RMS level error of the sum and of the tweeter, and largest PIR deviation, 100 Hz–16 kHz (dB).

	sum, RMS			tweeter, RMS			PIR		
d	10°	5°	2.5°	10°	5°	2.5°	10°	5°	2.5°
1 m BEM	0.62	0.41	0.38	0.96	0.63	0.58	0.55	0.29	0.15
1 m BEM, complex responses interpolated	1.21	0.47	0.37	1.81	0.70	0.58	0.63	0.36	0.15
1 m DED	0.35	0.26	0.28	0.57	0.43	0.45	0.08	0.05	0.02
2 m BEM	0.60	0.28	0.14	0.99	0.43	0.22	0.32	0.28	0.10
2 m BEM, complex responses interpolated	1.05	0.41	0.14	1.59	0.62	0.22	0.36	0.34	0.11
2 m DED	0.35	0.12	0.07	0.56	0.20	0.13	0.06	0.07	0.02
4 m BEM	0.58	0.27	0.09	0.94	0.42	0.15	0.16	0.16	0.07
4 m BEM, complex responses interpolated	1.02	0.50	0.12	1.52	0.75	0.17	0.20	0.20	0.10
4 m DED	0.31	0.11	0.03	0.52	0.19	0.06	0.04	0.04	0.02

Table 15: The vertical orbit measured with the cabinet on its side on the turntable: RMS level error, uncorrected and corrected with 10° and 5° steps (dB).

		sum			tweeter			woofer		
d	model	uncorr.	10°	5°	uncorr.	10°	5°	uncorr.	10°	5°
1 m	BEM	1.86	0.38	0.26	2.22	0.56	0.39	1.64	0.22	0.17
1 m	DED	1.88	0.19	0.12	2.30	0.33	0.21	1.69	0.12	0.05
2 m	BEM	1.00	0.30	0.17	1.19	0.42	0.24	0.87	0.14	0.07
2 m	DED	0.98	0.12	0.05	1.17	0.22	0.09	0.90	0.10	0.03
4 m	BEM	0.52	0.18	0.12	0.61	0.26	0.18	0.45	0.08	0.04
4 m	DED	0.50	0.08	0.03	0.60	0.13	0.06	0.46	0.07	0.03

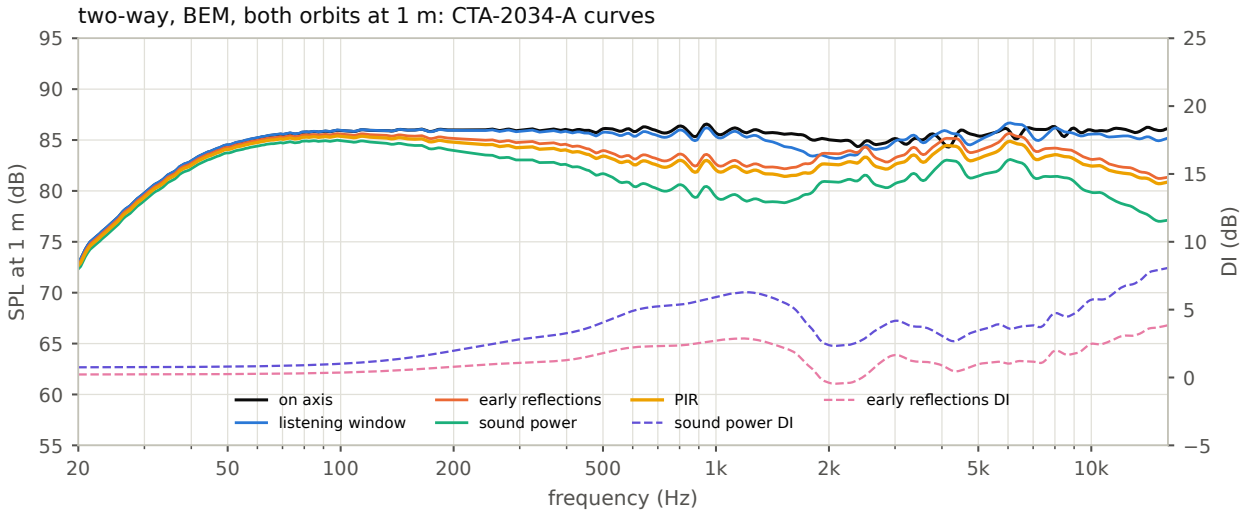


Figure 24: CTA-2034-A curves of the BEM two-way, rigorous setup, both orbits at 1 m.

the rigid dome sends more sound to the cabinet's edges than the DED model assumes, the polar patterns carry finer interference ripples, and with 10° steps the correction leaves about 1 dB RMS on the tweeter's individual curves at any distance, and up to about 0.55 dB on the PIR in narrow bands at 1 m (0.3 dB at 2 m). That residual is mostly the angular step: with 2.5° steps it falls to 0.2 dB RMS on the tweeter and 0.1 dB on the PIR at 2 and 4 m. At 1 m about 0.4 dB RMS remains on the sum whatever the step, the share of the edges' finite-distance geometry. The vertical orbit, measured with the cabinet on its side, corrects the same way. Rebuilt at a larger distance than measured, the data keep the cabinet's near field.

8 Discussion

The geometry is the easy part. For the direct sound of each driver the correction is exact up to interpolation, and the angular interpolation of a 10° orbit is good enough (about 0.1 dB RMS). The drivers' heights relative to the microphone do not matter at 1 m.

Diffraction is the limit, mostly behind the cabinet. The edge waves come from the baffle edges, not from the driver; the correction re-times them as if they did. In this simulation the consequence is small in front of the loudspeaker and grows behind it, at mid and high frequencies, where the edge waves are the response. Spatial averages are barely affected: the PIR of the corrected data is within about 0.1 dB of the rigorous one. Individual rear curves can show local deviations of a few dB in narrow bands. The boundary-element check (Section 7) shows that the DED model underestimates that diffraction for a dome tweeter: the rigid dome radiates wider, more sound reaches the edges, and the residual grows above 3 kHz, in front on the side of the far edge as well as behind — up to about 0.55 dB on the PIR in narrow bands at 1 m with 10° steps, 0.15 dB with 2.5° steps. Below 3 kHz the BEM confirms the DED results.

The angular step. With 10° steps the residual on individual curves stays near 0.6 dB RMS on the sum at every distance, because the interference patterns keep their angular period; 5° steps take a third off at 1 m and halve it at 2 and 4 m, 2.5° steps leave 0.38 dB at 1 m, 0.14 dB at 2 m and 0.09 dB at 4 m. Even the rigorous 10° data average the far edge's ripple at a few points of its period: their listening window differs by up to 0.6–0.7 dB from a dense average at 7–8 kHz.

Interpolate powers. Where the angular step undersamples the pattern, the two measurements that bracket a direction are not coherent and the linear interpolation of complex responses cancels in part, with artificial notches. Interpolating the re-propagated powers, keeping the phase of the complex interpolation, removes them; on the BEM data it halves the residual of the individual curves with 10° steps and changes nothing where the data are well sampled (2.5°). It has a small price on a well-sampled main lobe, where the mean of the powers reads slightly high: 0.1 dB RMS on the direct sound of the summed three-way instead of 0.03 dB, and 0.15 dB on its sound power.

The lateral offset. The model handles the laterally shifted microphone, provided the lateral offset is entered, with the right sign. Missing or with the wrong sign, it places the measurements up to 11° or 21° away from where they were taken at 1 m, and the individual curves keep two to seven times the residual error. With the powers interpolated, the spatial averages hardly notice; with the complex responses interpolated, the same mistakes made the PIR worse than no correction. A quick check: at +90° the microphone must face the side counted as positive, and the reference axis of a layout like this one lies on that side if the microphone moves toward the tweeter at positive angles.

1 m, 2 m, or 1 m rebuilt at 2 m. At 2 m the raw errors are halved and the finite-distance part of the diffraction error shrinks, but the angular-interpolation part does not; a finer orbit is then the

effective remedy. Rebuilding 1 m measurements as seen from 2 m fixes the drivers' geometry (a woofer beside the reference axis is seen from the right angles) but not the cabinet's near field: its low-frequency acoustic center ahead of the baffle and its diffraction, both different at 1 and 2 m. On the BEM two-way the rebuilt data keep about 0.65 dB RMS on the sum and 0.35 dB on the PIR, where 2 m measurements with 2.5° steps reach 0.14 and 0.06 dB. In a real room the choice is also about the reflection-free time window, which shrinks with distance — outside the scope of this free-field study.

Limits of the study.

- Free field. The room, the gating window and its interaction with the correction (the delays change by up to 1 ms behind the cabinet) are not simulated.
- The DED model is an approximation: only the edges of the front baffle are modelled, the field behind the cabinet is the sum of the edge dipoles, and the edges are sharp. The boundary-element check removes the first two approximations for a two-way variant and finds more diffraction above 3 kHz. Real cabinets with rounded edges diffract less, which should help the correction.
- Rigid pistons, cones and domes. Real diaphragms break up at high frequency and real tweeters sit in faceplates or waveguides, which change their directivity and how much sound reaches the edges. The acoustic centers are on the baffle plane: a real acoustic center behind or in front of the baffle acts like a depth error for that driver; ±10 mm of depth error costs little here, and the low-frequency acoustic center, 130 mm ahead of the baffle in the BEM, does not disturb the correction at the measurement distance.
- The vertical orbit, measured with the loudspeaker on its side (depth 145 mm, no lateral offset), was simulated for the two-way only (Table 15); the three-way's CTA-2034-A curves are horizontal-orbit estimates.

Checklist for a small-turntable polar measurement

1. Center the cabinet on the turntable; put the microphone on the reference axis (in front of the reference point) at 0°, at the reference height.
2. Measure the depth (rotation axis to baffle) and the lateral offset (rotation axis to reference axis) to within about 1 cm; note on which side the microphone faces at +90° and give the lateral offset that sign.
3. Give every driver its offset from the reference axis in the plane of rotation — a woofer beside the tweeter in a horizontal cabinet is not at 0 — and its height relative to the microphone.
4. Interpolate the powers between measured directions, not the complex responses.
5. Use the finest step you can afford if the responses above a few kHz matter: with this cabinet, 10° steps leave about 1 dB RMS on the tweeter's curves at any distance, 5° takes a third to a half off, 2.5° brings it to 0.2 dB at 2 m.
6. Measure at the distance you want the results at when the room allows it: rebuilt from 1 m, 2 m data keep the cabinet's near field.
7. Treat rear responses at mid and high frequencies as approximate (cabinet diffraction). Spatial averages (sound power, PIR, DI) are robust: within about 0.1 dB below 3 kHz, a few tenths of a dB in narrow bands above.

9 Conclusion

On a small turntable the cabinet is centered and the microphone moved sideways to face the reference point; uncorrected, the off-axis data are too low (up to 2.5 dB behind the cabinet at 1 m),

late and seen from the wrong direction, and the PIR is about 1 dB low. A geometric correction — each driver a point source with directivity, re-propagated in 3D and its powers interpolated — restores them: nearly exactly for the direct sound, and to within a few tenths of a dB RMS once the cabinet's edge diffraction is included, the residual being concentrated behind the cabinet at mid and high frequencies. The spatial averages of CTA-2034-A, PIR included, come within about 0.1 dB of the rigorous ones. A boundary-element solution of a two-way in the same cabinet confirms this below about 3 kHz; above, the dome tweeter drives the cabinet's edges harder than the edge model assumes, and the corrected PIR deviates by up to about 0.55 dB in narrow bands at 1 m (0.3 dB at 2 m) with 10° steps; 2.5° steps bring it to 0.15 dB at 1 m and 0.1 dB at 2 m. The lateral offset of the reference axis, with its sign, still decides how good the individual curves are. And a polar measurement is best taken at the distance it will be used at: rebuilt at another distance, it keeps the cabinet's near field.

A Additional figures

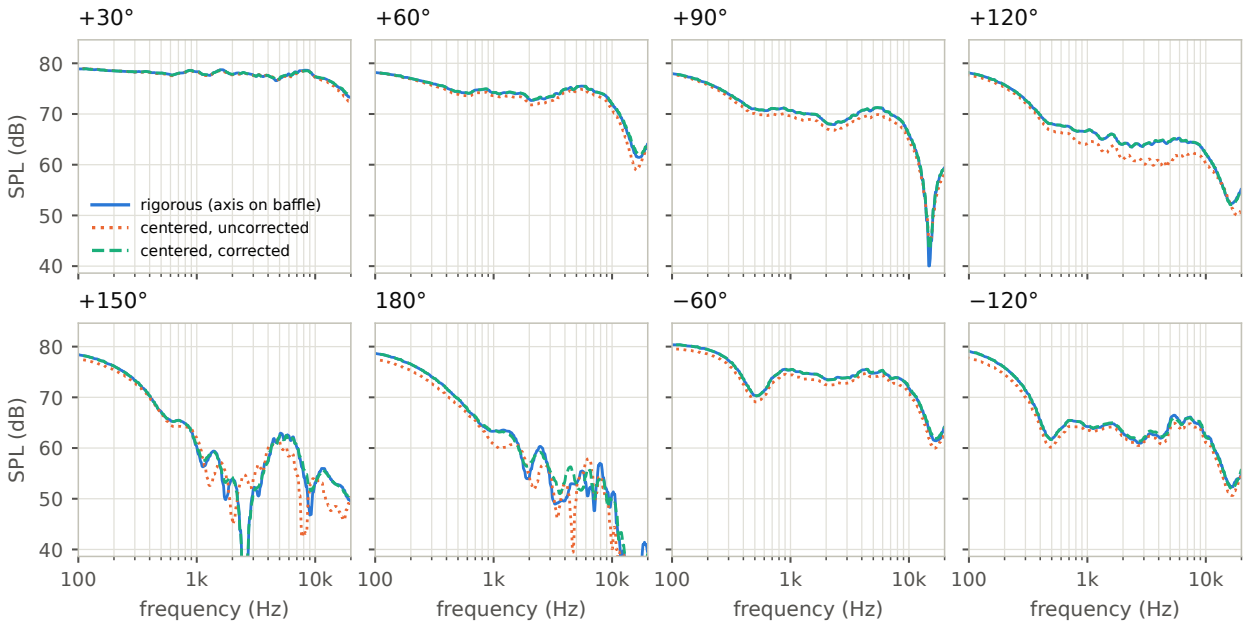


Figure 25: Summed loudspeaker at selected angles, 2 m (levels at 2 m), 1/12-octave smoothing.

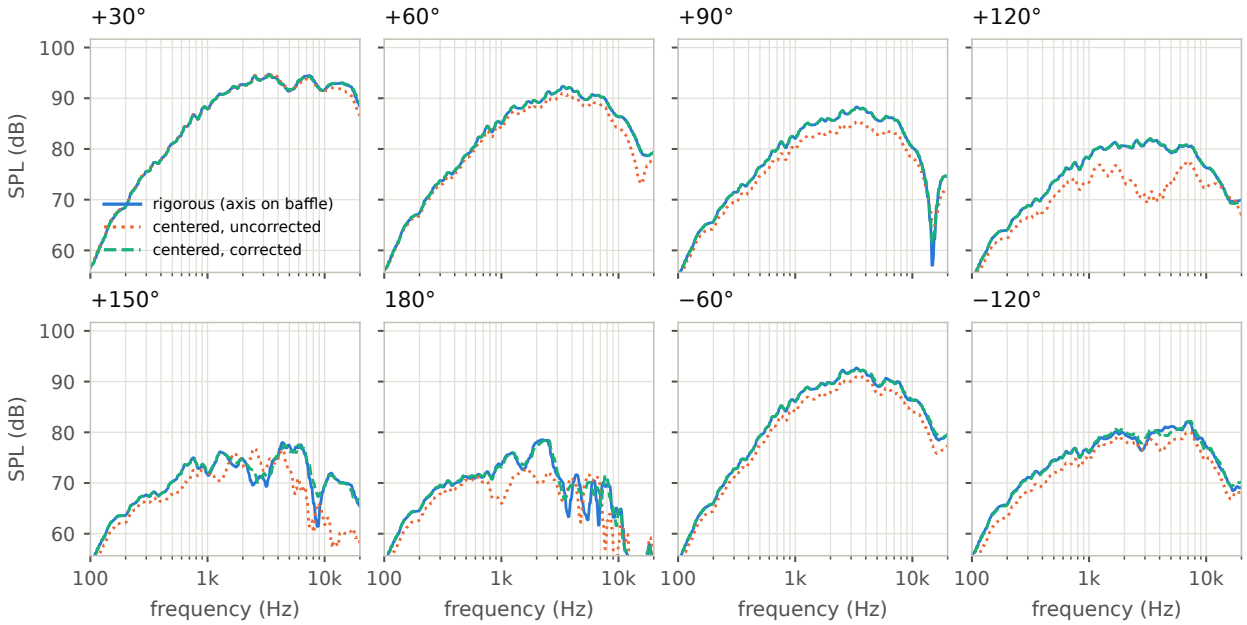


Figure 26: Tweeter alone at selected angles, 1 m, 1/12-octave smoothing. Behind the cabinet the response is made of edge waves; the corrected curves follow the rigorous ones with local deviations in the 4–8 kHz ripple.

B Reproducibility

Everything in this report is produced by the Python code of the repository (NumPy, SciPy, Matplotlib), from the boundary-element pressures for Section 7:

- `rotcenter/`: geometry and kinematics, radiators, driver models, DED diffraction, the correction (time and frequency domains), CTA-2034-A averages (one or both orbits), crossovers, metrics, the two-way variant;
- `scripts/run_study.py`: all simulated measurements of the three-way (≈ 1.5 min);
- `data/bem/`: the BEM pressures of the two-way (`two_way.npz`: each driver, per unit volume velocity, at the 1728 microphone positions of `points.json` and the 464 frequencies, with the mesh size and the time of each solve), and `scripts/import_bem.py`, which wrote it from the solver's report;
- `scripts/run_bem.py`: the two-way, BEM, DED and point sources, all cases (≈ 1 min);
- `scripts/make_figures.py` and `scripts/make_bem_figures.py`: figures and tables;
- `scripts/make_schematics.py`: the schematics;
- `tests/`: unit tests (kinematics, directivities, DED normalization, correction of an ideal point source in 3D and at another distance, the change of angle across the plane, the powers interpolated, symmetry copies, time and frequency domains, crossovers, CTA-2034-A curves, the two-way's microphone positions against the BEM data).

The boundary-element solver itself (Huygens) is not public.

The datasheet curves (`data/datasheets/`) were extracted from the vector graphics of the manufacturers' PDF datasheets; they belong to their manufacturers and are used here for analysis only.

References

[1] M. Urban, C. Heil, C. Pignon, C. Combet, P. Bauman, "The Distributed Edge Dipole (DED) Model for Cabinet Diffraction Effects," *J. Audio Eng. Soc.*, vol. 52, no. 10, pp. 1043–1059, 2004.

[2] ANSI/CTA-2034-A, *Standard Method of Measurement for In-Home Loudspeakers*, Consumer Technology Association, 2015.

[3] J. Vanderkooy, "A Simple Theory of Cabinet Edge Diffraction," *J. Audio Eng. Soc.*, vol. 39, no. 12, pp. 923–933, 1991.

[4] L. L. Beranek, T. J. Mellow, *Acoustics: Sound Fields and Transducers*, Academic Press, 2012.

[5] S. A. Sauter, C. Schwab, *Boundary Element Methods*, Springer, 2011.

[6] J. Vanderkooy, "The Low-Frequency Acoustic Center: Measurement, Theory, and Application," 128th Convention of the Audio Engineering Society, London, 2010.

[7] A. J. Burton, G. F. Miller, "The application of integral equation methods to the numerical solution of some exterior boundary-value problems," *Proc. R. Soc. Lond. A*, vol. 323, pp. 201–210, 1971.